

ENDOSOMAL SORTING COMPLEX REQUIRED FOR TRANSPORT machineries play crucial roles in vacuolar trafficking and cytokinesis in the moss *Physcomitrium patens*

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Summary

- The ENDOSOMAL SORTING COMPLEX REQUIRED FOR TRANSPORT (ESCRT) proteins are essential for converting early endosomes (EEs) into late endosomes/multivesicular bodies (LEs/MVBs) during endocytosis. In animals, they also play noncanonical roles in cytokinetic abscission, membrane repair, and other membrane remodeling processes. However, their functions in plants, especially nonvascular plants, remain poorly understood.
- Here, we systematically investigate the localization and functions of ESCRT proteins in the moss *Physcomitrium patens* using fluorescent protein labeling, mutant analyses, interaction assays, and electron microscopy.
- We show that ESCRT-related proteins localize to endosomes and play a conserved role in endocytic trafficking. Notably, all tested ESCRT components also localize to the developing cell plate, whereas reporters involved in *trans*-Golgi network (TGN)-MVB transition do not. Disruption of VPS4, a key regulator of ESCRT-III, delays cytokinesis progression and alters the dynamics of cell plate assembly factors.
- We propose that the ESCRT machinery plays both conserved and previously unrecognized noncanonical roles in plants. By providing this evidence in a bryophyte, our work establishes a critical reference point for future functional studies in nonvascular plants and for evolutionary comparisons across plant lineages.

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Introduction

Endosomal trafficking is essential for regulating membrane protein homeostasis and signaling (Arora & Damme, 2021; Gilleron & Zeigerer, 2023). In animals, internalized cargo is delivered into canonical early endosomes (EEs). Plants, however, have evolved a distinct mechanism, in which a dynamic subpopulation of the *trans*-Golgi network (TGN) serves as the primary destination for endocytosed materials (Dettmer *et al.*, 2006; Viotti *et al.*, 2010). Vesicles derived from the TGN/EE mature into late endosomes (LEs), characterized by the formation of intraluminal vesicles (ILVs). These LEs are also referred to as multivesicular endosomes (MVEs), multivesicular bodies (MVBs), or prevacuolar compartments (PVCs) (Tse *et al.*, 2004; Scheuring *et al.*, 2011). Ultimately, MVBs fuse with the vacuole to deliver their contents for degradation (Zeng *et al.*, 2023b).

In animals, cargo sorting and ILV formation are orchestrated sequentially by the Endosome Sorting Complexes Required for Transport complexes (ESCRT-0, -I, -II, and -III), and the AAA+ ATPase VPS4/SKD1 along with its activator LIP5/VTG1 (also referred to as ESCRT-IV) (Vietri *et al.*, 2020). These processes are coupled with progressive acidification of the endosomal

lumen, driven by RAB5 and RAB7 family GTPases (Podinovskaia & Spang, 2018; Ott *et al.*, 2025). In plants, almost all ESCRT components except ESCRT-0 are highly conserved and appear to perform analogous functions (Gao *et al.*, 2017; Gonzalez Solis *et al.*, 2022; Zeng *et al.*, 2023b; Weiner *et al.*, 2025). However, the subcellular localization of RAB GTPases in plants diverges markedly from their animal counterparts. Specifically, plant RAB5/RABF and RAB7/RABG primarily localize to the MVB and vacuolar membrane, respectively, rather than to EEs and LEs (Ueda *et al.*, 2001, 2004; Sohn *et al.*, 2003; Kotzer *et al.*, 2004; Lee *et al.*, 2004; Bottanelli *et al.*, 2012; Cui *et al.*, 2014; Ebine *et al.*, 2014; Singh *et al.*, 2014).

Endomembrane trafficking is also critical for plant cytokinesis (Fig. 1), during which a cell plate forms to separate daughter cells (Sinclair *et al.*, 2023). Secretory vesicles carrying cell wall materials are transported to the developing cell plate by kinesin-12II motors (Reichardt *et al.*, 2007; Chow *et al.*, 2008; Sinclair *et al.*, 2023; Yamada *et al.*, 2025). These vesicles undergo homotypic fusion and are subsequently remodeled by dynamin GTPases into a tubular-vesicular network (Samuels *et al.*, 1995; Otegui *et al.*, 2001; Kang *et al.*, 2003; Segui-Simarro *et al.*, 2004). This network then transitions into a planar

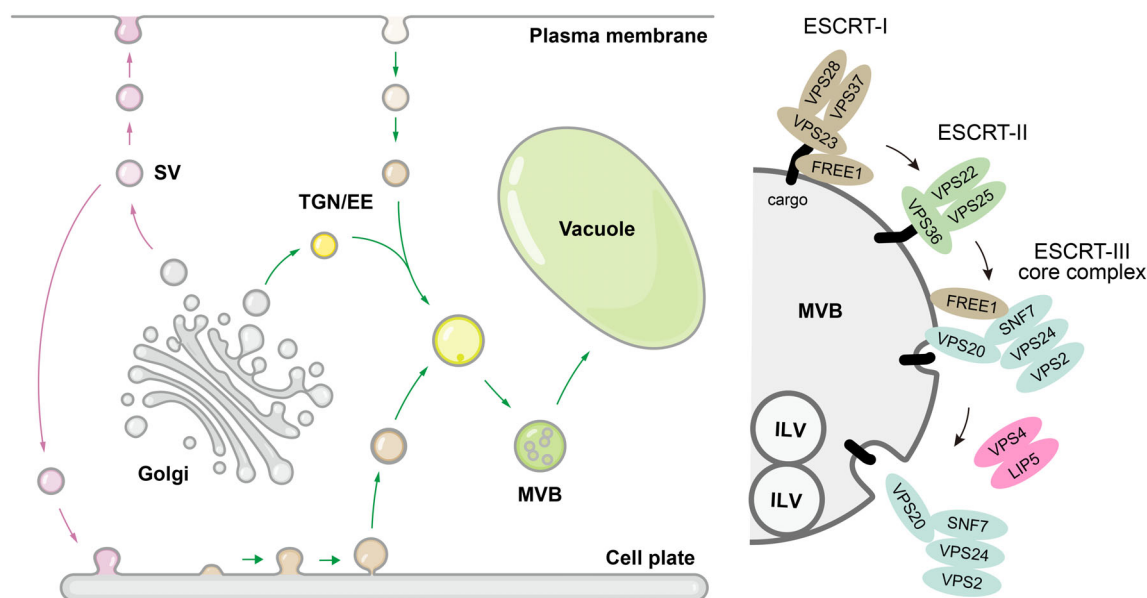


Fig. 1 Simplified model of endomembrane trafficking in plants. In the secretory pathway, secretory vesicles (SVs) bud from the *trans*-Golgi network (TGN) and are delivered to the plasma membrane during interphase and to the cell plate during cytokinesis. The TGN also functions as an early endosome (EE), receiving internalized cargos and maturing into late endosomes/multivesicular bodies (LEs/MVBs). During cytokinesis, some membrane components from the cell plate are endocytosed and delivered to LEs/MVBs. MVB biogenesis is driven by the sequential action of the endosomal sorting complexes required for transport (ESCRT)-I, -II, -III complexes together with VPS4 and LIP5, through which ubiquitinated cargos are sorted into intraluminal vesicles (ILVs) via ESCRT-III-mediated membrane invagination. For simplicity, the ESCRT-III accessory complex and other ESCRT-related components are not shown.

fenestrated sheet and eventually a mature cell plate (Sinclair *et al.*, 2023). Beyond secretion, endocytosis also contributes to plant cytokinesis, though its precise role remains debated (Samuels *et al.*, 1995; Otegui *et al.*, 2001; Segui-Simarro *et al.*, 2004; Baluska *et al.*, 2005; Dhonukshe *et al.*, 2006; Segui-Simarro & Staehelin, 2006; Spitzer *et al.*, 2006; Reichardt *et al.*, 2007; Tahara *et al.*, 2007; Shahriari *et al.*, 2010; Van Damme *et al.*, 2011; Song *et al.*, 2012; Richter *et al.*, 2014). Some evidence suggests that endocytosed materials are redirected to the cell plate to support its construction (Baluska *et al.*, 2005; Dhonukshe *et al.*, 2006; Richter *et al.*, 2014), whereas others favor a model that endocytosis recycles cell plate assembly factors, such as Qa-SNARE SYP111/KNOLLE, for degradation via the canonical MVB-vacuole pathway (Reichardt *et al.*, 2007; Chow *et al.*, 2008).

The ESCRT machinery is evolutionarily conserved in eukaryotes. Beyond its central role in MVB maturation, this machinery broadly participates in membrane remodeling processes in animals, including cytokinetic abscission, nuclear envelope repair, virus budding, and autophagosome closure (Vietri *et al.*, 2020). Some of these functions, such as autophagosome closure (Zeng *et al.*, 2023a), are conserved in plants. However, our understanding of ESCRT functions in plants derives largely from studies in *Arabidopsis*, leaving it unclear whether noncanonical, plant-specific roles exist and how these functions have diverged across land plants. Here, we report that ESCRT components in the moss *Physcomitrium patens* (*P. patens*) play conserved roles in MVB biogenesis and vacuolar trafficking. Unexpectedly, they also localize to the cell plate during cytokinesis and regulate this

process via a mechanism that appears to operate independently of the TGN.

Materials and Methods

Moss strains and culture conditions

The moss strains used in this study were derivatives of the Grandson *Physcomitrium patens* (formerly *Physcomitrella patens*) (Engel, 1968), with their genotypes detailed in Supporting Information Table S1. Moss plants were cultured on BCDAT plates under continuous light at 24°C. To analyze colony growth, nearly equal amounts of protonema tissue were transferred to BCDAT plates and grown for 14 d. Colony images were captured using a digital camera at 2-d intervals. For high-resolution imaging, moss samples were grown for 5–7 d on a thin layer of BCD medium in 35-mm bottom-glass imaging dishes.

Sequence and phylogenetic analyses

Moss homologs of proteins in this study were identified by performing BLAST searches using *Arabidopsis* counterparts against the *P. patens* proteome (v.6.1) in Phytozome (<https://phytozome-next.jgi.doe.gov>). Sequence alignments were conducted using the CLUSTALO program integrated in the JALVIEW software (<https://www.jalview.org/>). Maximum-likelihood phylogenetic trees were constructed using the FASTTREE program with default settings on the Galaxy platform (<https://usegalaxy.org/>) and visualized using iTOL (<https://itol.embl.de/>).

Plasmid construction

Knockin plasmids for VPS4A, LIP5, VPS28A, VPS20, DRP1A, and GNOM3 were constructed by simultaneously ligating three fragments into the pPY22 vector (<https://www.addgene.org/137082/>) via the In-Fusion Snap Assembly Kit (cat. 638948; Takara Bio, Dalian, China). The fragments included a *c.* 1 kb 5' homologous fragment upstream of the stop codon, a *c.* 3.5 kb in-frame fluorescent protein-coding fragment followed by the *nos* terminator and an antibiotic resistance selection cassette (Hygromycin B or Geneticin resistant), and a *c.* 1 kb 3' homologous fragment downstream of the stop codon. Knockout plasmids for *vps4a* and *vps4b* were constructed similarly to the knockin plasmids, with the following modifications: the *c.* 1 kb 5' homologous fragment was amplified from the genomic sequence upstream of the start codon, and an antibiotic resistance cassette (Hygromycin B or Nourseothricin resistant) without a fluorescent protein-coding sequence was inserted between the 5' and 3' homologous fragments. Plasmids for the constitutive expression of proteins were constructed for localization analysis and rescue experiments. The coding sequences were inserted into a pENTR vector containing an in-frame N-terminal (for VSR1, VSR1 (TMD-CT), RABF2A, and RABA1D) or C-terminal (for VPS4A, VPS4B, VPS20, VPS36, RABF1A, STX6A, VHA-A1, AtSKD1, ScVPS4, VPS4B(ΔMIT), and VPS4B(MIT)) fluorescent protein-coding sequence (mNeonGreen, mStayGold(J), mCherry, or mYonghong) using In-Fusion ligation. The mNeonGreen sequence was derived from pPY22 (Yi & Goshima, 2020a). The sequences of mStayGold(J) and mYonghong were obtained from the literature (Shimozono *et al.*, 2024; Xiong *et al.*, 2025), codon-optimized, and synthesized by Tsingke Biotech. The resultant fluorescent fusion fragments in pENTR plasmids were transferred into a destination vector via LR reactions (LR Clonase Enzyme Kit, cat. 11791019; Thermo Fisher Scientific, Shanghai, China), enabling expression under the control of the EF1 α promoter. The destination vector contained an antibiotic resistance cassette (Hygromycin B or Geneticin resistant) and homologous recombination sites (HB7 or PTA1 locus) for gene targeting. The inducible expression plasmid for VPS4A(E237Q) was constructed by inserting the VPS4A(E237Q)-Cerulean coding sequence into the β -estradiol-inducible expression vector (pPGX8) via an LR reaction. To achieve mitotic-specific inducible expression, the KIN1D1a promoter in the pPGX8 vector was replaced with a 1999 bp fragment consisting of the CYCA2 promoter (1682 bp, Pp6c2_4910) and its N-terminal destruction box (609 bp), creating a vector for mitotic expression of LexA-BD:VP16:ER-LBD. Subsequently, the coding sequence of VPS4B(E236Q)-mNG fusion fragment was cloned into this modified vector via an LR reaction. Single guide RNA (sgRNA) targets for generating *vps4a*(M312T/L332S) mutants were manually selected: GCTCGGCAGCACATGTTTA (AGG) and CGATTATGAGGA TCTGGCC (AGG). These targets were inserted into the sgRNA expression vector pPY156 via PCR-based site-directed mutagenesis. Single-stranded oligonucleotide templates for introducing the M312T and L332S mutations were commercially

synthesized: 5'-GTTTAAGGCTCGGC AGCACACGTTTAAA GTGCTCATGGTGTTCATAT-3' (oPY550) and 5'-CTGA ACGCGATTATGAGGATAGTGCACGGAAGACTGATGG ATTTTCA-3' (oPY551). All plasmids and primers used in this study are listed in Tables S2 and S3, respectively.

Moss transformation

Moss protonemata were harvested and enzymatically digested using 0.5% (w/v) driselase to isolate protoplasts. The protoplasts were then transformed with 30 μ g of linearized plasmids to facilitate homologous recombination for gene knockout, knock-in, and constitutive expression using the PEG-mediated transformation protocol. For CRISPR/Cas9-mediated knock-in of *vps4a*(M312T/L332S), two sequential transformations were performed. Each transformation used 10 μ g of Cas9-expressing plasmid, 10 μ g of an sgRNA plasmid, and 10 μ l of 50 μ M single-stranded oligonucleotide to introduce the M312T and L332S mutations into *vps4b* knockout mutants, respectively. Following transformation, the protoplasts were plated on PRM plates overlaid with cellophane and cultured for 10–14 d. During this period, the cellophane disks were transferred to selection plates: once for transient selection of CRISPR/Cas9-mediated knockin and knockout lines, and twice for the selection of homologous recombination-mediated gene targeting lines. Positive colonies were propagated and verified by PCR amplification and Sanger sequencing using KOD-ONE DNA polymerase (cat. KMM-101; Toyobo, Shanghai, China).

Imaging and data processing

Images were acquired by using the CELLSENS DIMENSION software operating on the Olympus SpinSR microscopy system. The system included an inverted microscope (IX83P2ZF) equipped with 10 $\times$ (NA 0.40), 20 $\times$ (NA 0.70), 40 $\times$ (NA 0.95), 60 $\times$ (NA 1.42), or 100 $\times$ (NA 1.48) objectives, a spinning disk confocal scanner (Yokogawa CSU-W1) connected to 405, 488, and 561-nm laser lines, and an ORCA Flash 4.0 V3 camera (Hamamatsu). Time-lapse imaging was performed at intervals of 1–5 min for visualizing cell division or 0.5–2 s for capturing subcellular dynamics. The laser power and exposure time ranged between 20–50% and 200–500 ms, respectively. Fluorescence recovery after photobleaching experiments were carried out using the FV3000 scanning module. Bleaching was stimulated by a 20% 561-nm laser for 5 s. Images were taken by a 5% 561-nm laser at an interval of 0.5 s using the Fv31s-sw software. Image processing and data extraction were conducted using Fiji software (v.2.14.0). Kymographs in Fig. 3(f) were generated by drawing a line along the division plane and reslicing the time-lapse images. Intensity plots in Fig. 3(h) were derived from a representative vertical line region in kymographs of Fig. 3(g). Intensity plots in Fig. 3(j) were generated from a representative line region in Fig. 3(i) along the division plane. The appearance time and duration of VPS4A and LIP5 on the cell plate in Fig. 3(h) were measured from vertical distances in the corresponding kymographs in Fig. 3(g). Cytokinesis duration in Fig. 6 was measured from the onset of phragmoplast expansion to its complete disassembly.

Drug treatment

For transient treatment with wortmannin, latrunculin A, and oryzalin, mosses were cultured in 35-mm imaging dishes for 5–7 d and preincubated in 2 ml sterile water for 30 min. Subsequently, 100 μ l wortmannin, latrunculin A, or oryzalin was added to achieve a final concentration of 6, 5, and 5 μ M, respectively. Samples were incubated for 5–30 min at room temperature before imaging. Because higher concentrations of wortmannin frequently induced cell death, 6 μ M was determined to be the optimal concentration for inducing MVB enlargement without significantly affecting cell viability within 1 h. For long-term treatment with BFA, moss plants were cultured in BCD medium supplemented with 10 μ M BFA for 5–7 d. For FM4-64 treatment, 100 μ l of 10 μ M FM4-64 diluted in sterile water was added directly to mosses cultured in 35-mm imaging dishes. Samples were incubated in the dark for 5–60 min. Before imaging, the FM4-64 solution was removed. For transient β -estradiol treatment, imaging samples were treated with 100 μ l of 1 μ M β -estradiol for 5 min or longer at room temperature. The β -estradiol solution was removed before imaging. For long-term β -estradiol treatment, moss plants were cultured in BCDAT medium supplemented with 1 μ M β -estradiol for 9 d.

Yeast two-hybrid assay

Approximately 0.1 μ g of plasmids encoding AD- or BD-fusion proteins, along with 100 μ g of carrier DNA, were co-transformed into competent AH109 yeast cells. The transformed cells were plated onto solid YPDA synthetic dropout medium lacking Trp and Leu (2SD medium) for selection. Positive colonies were isolated and transferred to solid YPDA synthetic dropout medium devoid of Trp, Leu, His, and Ade (4SD medium) to assess protein–protein interactions.

Transmission electron microscopy

Mosses cultured in 35-mm imaging dishes were randomly collected and fixed in 2.5% glutaraldehyde. Subsequent processing and TEM were conducted by service from Sichuan Scientist Biotechnology. Briefly, fixed tissues were washed in 0.1% phosphate buffer (pH 7.4) three times, fixed again in 1% osmium tetroxide for 2 h, and dehydrated through an ethanol gradient from 50% to 100% and 100% acetone. Tissues were then embedded in EMBed 812 resin (90529-77-4; SPI Supplies, West Chester, PA, USA) and thin-sectioned into 60–80-nm slices by an ultramicrotome (EM UC7 equipped with a Diatome Ultra 45° diamond cutter; Leica, Shanghai, China). Slices were stained in 2% uranyl acetate and lead citrate, each for 15 min, dried overnight at room temperature, and imaged under an electron microscope (HT7800; Hitachi, Shanghai, China).

Statistical analysis

Data are presented as mean $\pm$ SD or box/violin plots with median and range (minimum–maximum, interquartile) displayed. Sample sizes are indicated in the figure legends or the main text.

Statistical comparisons were performed using appropriate tests shown in the figure legends. A *P*-value of less than 0.05 was considered statistically significant.

Results

ESCRT components are conserved in the moss *P. patens* and localize to endosomal compartments

To explore the functions of ESCRT proteins, we used the moss *P. patens* as a model because it is exceptionally well suited for cell biology studies (Rensing *et al.*, 2020). The *P. patens* genome encodes almost all ESCRT proteins, including the plant-specific subunits FREE1/FYVE1 and FYVE4 (Table S4) (Gao *et al.*, 2014, 2017; Liu *et al.*, 2021; Bi *et al.*, 2024). Publicly available transcriptomics data indicate that all ESCRT components are broadly expressed in major tissues (Ortiz-Ramirez *et al.*, 2016), albeit expression levels vary considerably among individual members (Fig. S1). We first generated mNeonGreen (mNG)-fusion knock-in or overexpression lines for VPS28A (ESCRT-I), VPS36 (ESCRT-II), VPS20 (ESCRT-III), VPS4A/SKD1, and LIP5/VTG1 for localization analysis (Fig. 1). As shown in Fig. 2(a), VPS28A, VPS4A, and LIP5 labeled punctate structures and were absent from other membranous compartments. Time-lapse imaging revealed dynamic movement and fusion events in VPS4A-labeled compartments (Fig. 2b). VPS36 predominantly localized to the cytosol and plasma membrane, with only occasional puncta detected (Fig. 2a). VPS20 showed weak plasma membrane association and labeled numerous oval vesicles (Fig. 2a). These vesicles were larger than VPS4A- and LIP5-labeled puncta, whereas no significant difference was observed between VPS4A- and LIP5-labeled structures (Fig. 2c). Overexpressed VPS4B fused with mCherry displayed a similar localization pattern to that of VPS4A and partially overlapped with VPS20-positive compartments (Fig. 2d). The diverged localization patterns of ESCRT components may be common in land plants, as similar findings have been reported for VPS28 (Scheuring *et al.*, 2011; Liu *et al.*, 2020), VPS36 (Wang *et al.*, 2017), VPS20 (Cai *et al.*, 2014), VPS4 (Haas *et al.*, 2007; Shahriari *et al.*, 2010), and LIP5 (Wang *et al.*, 2015; Buono *et al.*, 2016) in Arabidopsis.

Interestingly, punctate structures labeled by VPS28A, VPS20, VPS4A, and LIP5 all closely associated with actin foci labeled by Lifeact-mCherry (Fig. 2e). This association was further confirmed by time-lapse imaging of VPS4A-mNG and Lifeact-mCherry (Fig. 2f). When cells were treated with 5 μ M latrunculin A (LatA), an inhibitor that effectively blocks actin filament assembly while maintaining cell viability (Yi & Goshima, 2020a), the motility of VPS4A-positive puncta was significantly impaired, as evidenced by a marked reduction in the area occupied by these puncta over time (Fig. 2g,h). Such effects were not observed for the microtubule inhibitor oryzalin (Fig. S2). This suggests that these puncta were not protein aggregates but rather organelles, likely endosomes, whose motility depends on actin filaments but not microtubules. Accordingly, upon staining with the lipid dye FM4-64, VPS28A, and VPS4A

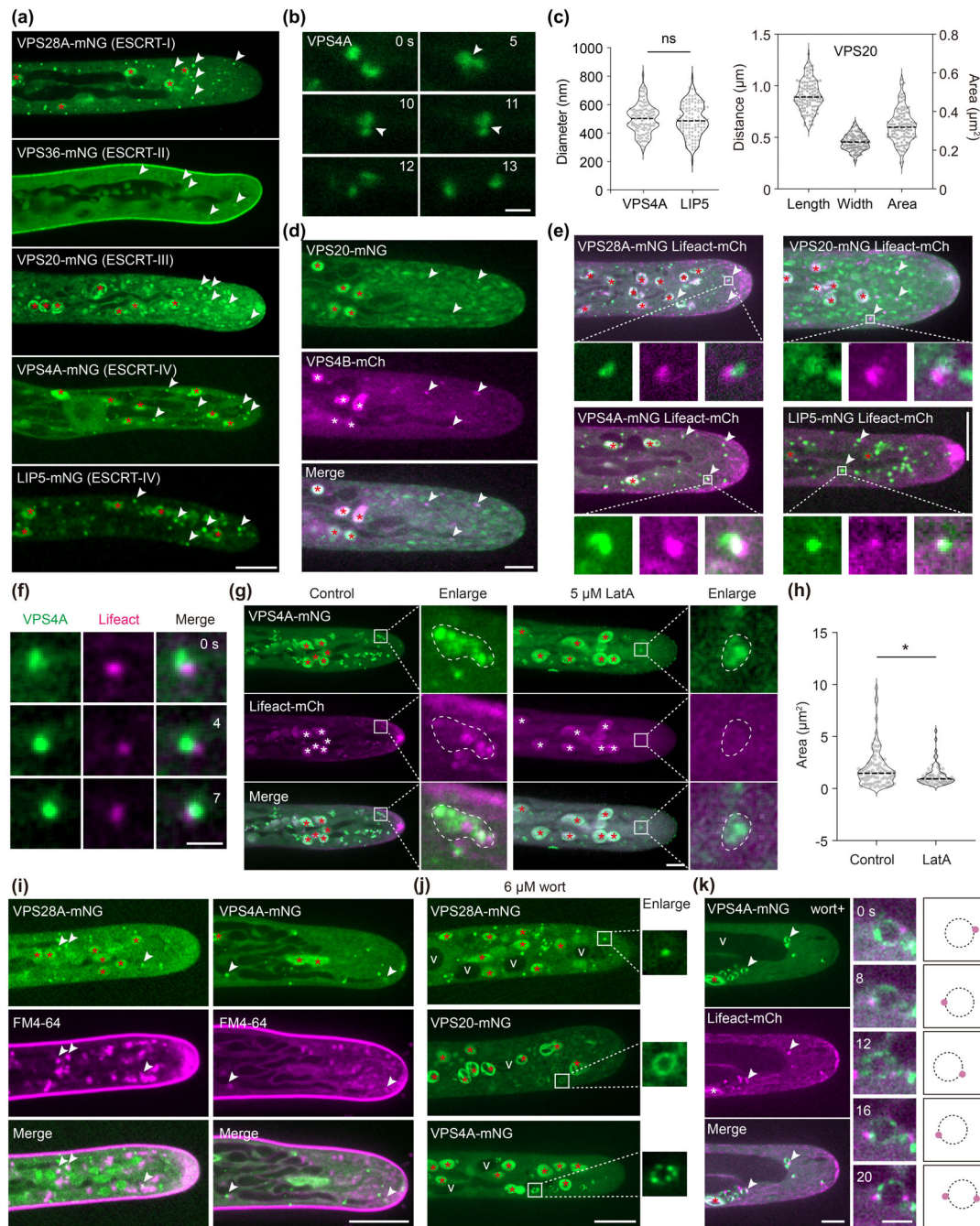


Fig. 2 Endosome Sorting Complexes Required for Transport (ESCRT)-related components localize to endosomal compartments. (a) Localization of ESCRT-I (VPS28A), ESCRT-II (VPS36), ESCRT-III (VPS20), and VPS4/LIP5 in *Physcomitrium patens* caulonemal tip cells. Each protein was fused with a C-terminal mNeonGreen (mNG) tag at its endogenous genomic site through homologous recombination. Arrowheads indicate vesicular structures. Asterisks mark chloroplast autofluorescence in (a, d, e, g, i–k). Bar, 10 μ m. (b) Vesicular structures labeled by VPS4A exhibit dynamic movement and transient associations (arrowheads). Bar, 1 μ m. (c) Quantification of vesicle size labeled by VPS4A, LIP5, and VPS20. VPS4A, $n = 111$; LIP5, $n = 109$; VPS20, $n = 108$. Violin plots show the data distribution, individual data points, minimum/maximum values, and the median (dashed line). ns, not significant (Student's t -test). (d) Partial colocalization of VPS20 and VPS4B (arrowheads). VPS4B was fused with a C-terminal mCherry (mCh) tag and expressed under the control of the EF1 α promoter. Bar, 10 μ m. (e) Close association between vesicles marked by ESCRT components and actin foci (arrowheads). Actin filaments were labeled with Lifeact-mCherry. Representative colocalized regions are enlarged below. Bar, 10 μ m. (f) Dynamic association between VPS4A-labeled puncta and actin foci. Bar, 1 μ m. (g) Disruption of actin assembly with latrunculin A (LatA) reduces the motility of VPS4A-labeled puncta. Maximum intensity projections from a 50-s movie are shown. Representative regions are enlarged. Bar, 10 μ m. (h) Quantification of the area occupied by VPS4A-labeled puncta. Control: $n = 89$; LatA, $n = 59$. Violin plots show the data distribution, individual data points, minimum/maximum values, and the median (dashed line). *, $P < 0.05$ (Mann–Whitney test). (i) VPS28A and VPS4A localize to FM4-64-labeled endosomes (arrowheads). Bar, 10 μ m. (j) Wortmannin induces enlargement of vesicles labeled by VPS20 and VPS4A, but not by VPS28A. Note that VPS4A forms discrete puncta on the vesicular membrane. Representative regions are enlarged. v, vacuole. (k) VPS4A-positive vesicles enlarged by wortmannin associate with actin foci (arrowheads). Bars: left, 10 μ m; right, 1 μ m.

showed partial colocalization with endocytosed FM4-64 (Fig. 2i). We further treated plants with wortmannin, a phosphoinositide 3-kinase (PI3K) inhibitor known to induce homotypic fusion and enlargement of MVBs (Wang *et al.*, 2009). Because higher concentrations of wortmannin rapidly caused cell death, we used 6 μ M as the optimal concentration. Intriguingly, the morphology of VPS28A-positive puncta was unaffected by wortmannin treatment, whereas VPS20- and VPS4A-labeled puncta were markedly enlarged (Fig. 2j). Additionally, VPS4A formed discrete dots on enlarged compartments, often accompanied by one or two actin foci (Fig. 2k). Together, these findings indicate that ESCRT components in *P. patens* localize to distinct subpopulations of endosomal compartments.

ESCRT-related proteins are dynamically recruited to the cell plate during cytokinesis

In Arabidopsis, loss of function of VPS23A and VPS4 leads to a multinucleate phenotype, suggesting a potential role of ESCRT proteins in cell division (Spitzer *et al.*, 2006; Shahriari *et al.*, 2010). To investigate whether ESCRT proteins participate in cell division in *P. patens*, we analyzed the dynamic localization of VPS28A, VPS36, VPS20, VPS4A, and LIP5 using time-lapse imaging. Unexpectedly, all these components were recruited to the developing cell plate during cytokinesis (Fig. 3a–e; Videos S1–S5). Their fluorescence signals first appeared in the cell center and subsequently expanded to the entire cell plate. Noticeably, VPS28A, VPS4A, and LIP5 formed discrete puncta on the cell plate, and their signals were rapidly depleted in an inside-out manner. The local enrichment and depletion of VPS20 and VPS36 were less pronounced. VPS4A- and LIP5-labeled puncta remained static throughout cytokinesis, with individual spots continuously emerging and disappearing (Fig. 3f; Videos S6, S7). The lifetime of each spot ranged from several to tens of seconds.

We next analyzed the timing of VPS4A and LIP5 recruitment relative to phragmoplast assembly, a microtubule-actin structure that guides cell plate formation and expansion. As shown in Fig. 3(d,e), both reporters were recruited when the actin-labeled phragmoplast began to expand. Quantitative analysis revealed an *c.* 15-min lag in the appearance of VPS4A and LIP5 following actin assembly (Fig. 3g,h). The late recruitment of VPS4A to the division plane was further validated using a microtubule reporter (Fig. S3a,b). Although actin regulates the motility of VPS4A-labeled compartments, it did not affect the localization of VPS4A on the cell plate (Fig. S3c). Therefore, VPS4A and likely other ESCRT proteins may not be delivered to the cell plate via actin-dependent transport.

Because ESCRT complexes assemble in a hierarchical manner, we analyzed colocalization between VPS20 and VPS4A to investigate their spatiotemporal relationship. As expected, VPS20 occupied a broader region of the cell plate than VPS4A, indicating its earlier recruitment (Fig. 3i). Moreover, VPS20 partially colocalized with VPS4A-positive puncta (Fig. 3j), suggesting that they may reside in the same compartments. RAB5 GTPases are well-established molecular switches that regulate endosomal trafficking and MVB biogenesis (Ueda *et al.*, 2001, 2004; Sohn

et al., 2003; Kotzer *et al.*, 2004; Lee *et al.*, 2004; Goh *et al.*, 2007; Bottanelli *et al.*, 2011, 2012; Ebine *et al.*, 2011; Jia *et al.*, 2013; Hao *et al.*, 2023). In Arabidopsis, RAB5s have been reported either to localize to the cell plate or to be absent from it (Dhonukshe *et al.*, 2006; Reichardt *et al.*, 2007). This prompted us to ask whether RAB5 also participates in cytokinesis in *P. patens*. Land plants possess two types of RAB5s: a canonical subfamily (RABF2) carrying a C-terminal prenylation motif, and a plant-specific subfamily (RABF1) with an N-terminal myristoylation signal (Fig. 4a). We selected RABF1A and RABF2A for localization analysis due to their relatively higher expression in caulonemal cells (Fig. S4). To avoid membrane-targeting interference, each protein was fused with an mCherry tag at the N-terminus (for RABF2A) or C-terminus (for RABF1A). As shown in Fig. 4a, RABF1A and RABF2A exhibited similar localization patterns: both were enriched at the plasma membrane and localized to oval vesicles resembling those labeled by VPS20 (Videos S8, S9). During cytokinesis, RABF1A and RABF2A were recruited to the cell center and then expanded to the entire cell plate, mirroring the behavior of VPS20 (Fig. 4b,c; Videos S10, S11). Their recruitment to the cell plate clearly preceded that of VPS4A (Fig. 4b,c). Together, these findings suggest that ESCRT-related proteins are recruited to the cell plate in a step-wise fashion.

ESCRT localization at the cell plate may not involve local TGN–MVB transition

The cell-plate localization of ESCRT proteins raises two possibilities: either MVBs undergo local maturation at the cell plate, or these proteins participate in other membrane remodeling processes. In Arabidopsis, MVBs mature directly from TGN/EE, as evidenced by extensive colocalization between TGN reporters and ESCRT-I (VPS28) and gradual loss of TGN markers on ESCRT-II/-III-positive MVBs (Scheuring *et al.*, 2011). If local maturation occurs at the cell plate, the TGN would be expected to accumulate there. To test this, we examined the localization of three well-established Arabidopsis TGN markers: VHA-A1, SYP61, and VSR1. VHA-A1 encodes a V-ATPase subunit required for MVB acidification (Dettmer *et al.*, 2006); SYP61 is a Qc-SNARE involved in membrane fusion during TGN-to-MVB/vacuole trafficking (Sanderfoot *et al.*, 2001); and VSR1 functions as a receptor for sorting vacuolar cargos from the TGN to endosomes and the vacuole (Sanderfoot *et al.*, 1998; Li *et al.*, 2002). The *P. patens* genome encodes five VHA-A1 homologs (VHA-A1 to VHA-A5), two SYP61 homologs (STX6A and STX6B), and eight VSR1 homologs (VSR1 to VSR8). Based on transcriptomic data (Ortiz-Ramirez *et al.*, 2016), we selected the homolog with the highest expression in caulonemal cells for each category – VHA-A1, STX6A, and VSR1 – for localization analysis. Given that SYP61 exhibits similar localization patterns when fused to either N- or C-terminal fluorescent tags (Luo *et al.*, 2015; Jones *et al.*, 2017), we used the C-terminal tagging strategy to facilitate vector construction for STX6A.

Unlike its TGN localization in Arabidopsis, moss VHA-A1 fused with mNG localized primarily to the tonoplast, with only

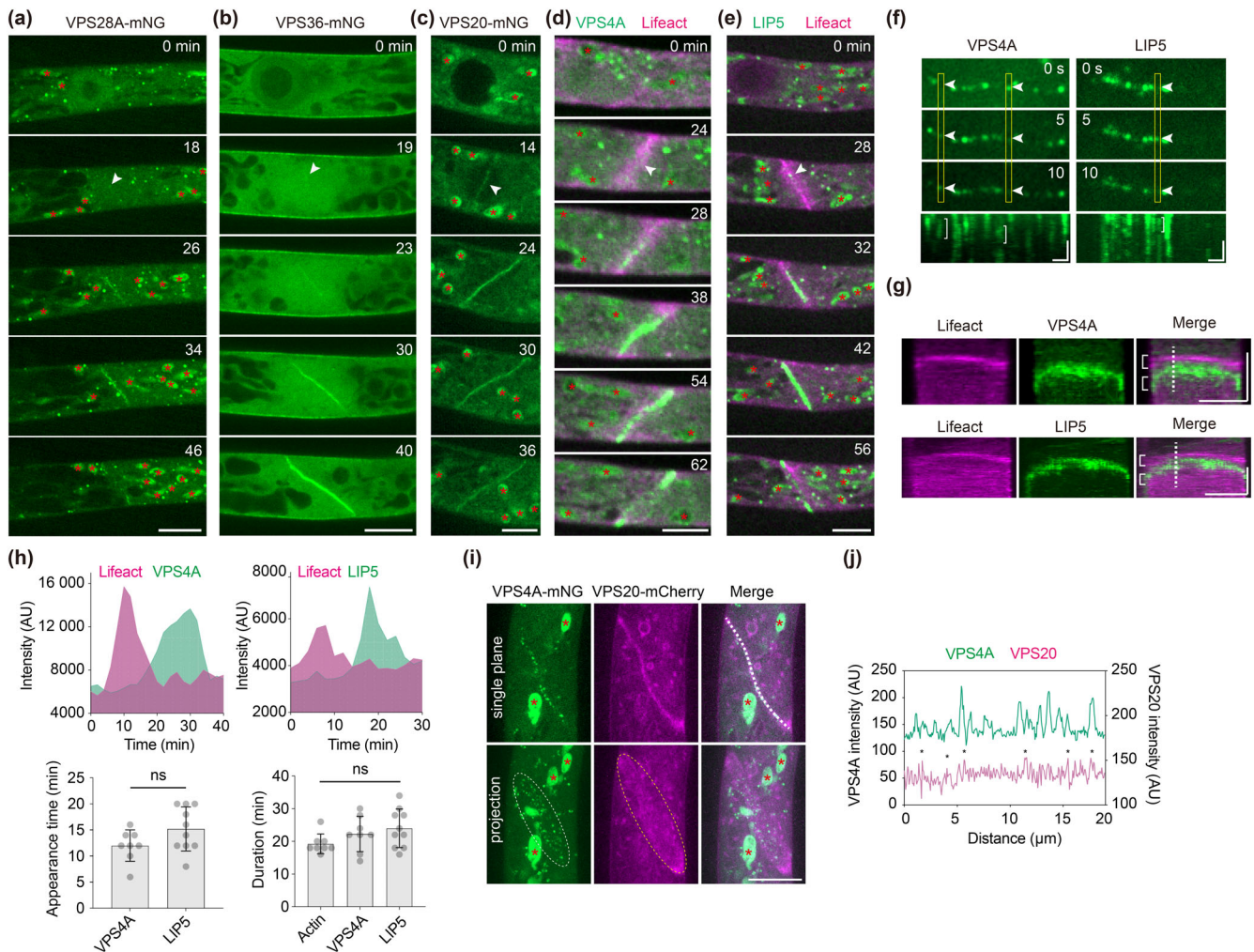


Fig. 3 Endosome Sorting Complexes Required for Transport (ESCRT)-related components localize to the cell plate during cytokinesis. (a–e) VPS28A (a), VPS36 (b), VPS20 (c), VPS4A (d), and LIP5 (e) are recruited to the cell plate in *Physcomitrium patens* caulonemal tip cells. All reporters first arise at the midline (arrowheads) and then expand to the entire cell plate. Asterisks indicate chloroplast autofluorescence in (a, c–e, i). Bars, 10 μ m. (f) VPS4A and LIP5 signals remain static on the cell plate. The emergence and disappearance of representative puncta are indicated by yellow boxes, arrowheads, and brackets. Bars: horizontal, 1 μ m; vertical, 10 s. (g) Kymographs drawn across the division plane show the appearance of actin and VPS4A/LIP5. Brackets denote the overall residence time of signals on the cell plate. Bars, horizontal, 10 μ m; vertical, 1 h. (h) Representative intensity plots for VPS4A, LIP5, and actin over time, generated from the dashed lines in (g). The appearance time of VPS4A and LIP5 relative to actin was measured as the distance between intensity peaks. The duration of each reporter was measured within the bracket-marked regions in (g). VPS4A (left), $n = 8$; LIP5 (left), $n = 10$; actin (right), $n = 8$; VPS4A (right), $n = 8$; LIP5 (right), $n = 10$. Quantitative data are presented as mean $\pm$ SD, with individual data points shown. ns, not significant; left, Student's t -test; right, one-way ANOVA. (i) Colocalization of VPS4A and VPS20 on the cell plate. Note that VPS20 marks a larger area than VPS4A. Bar, 10 μ m. (j) Representative intensity plots for VPS4A and VPS20 on the cell plate, drawn from the dashed line in (i). Asterisks indicate overlapping puncta labeled by both reporters.

occasional signals on vesicular compartments (Fig. S5a,b). STX6A and VSR1, tagged with mNG and mYonghong (mYH) (Xiong *et al.*, 2025), respectively, both labeled punctate structures and were largely colocalized (Fig. 4d,g). Additionally, VSR1 showed weak signal at the tonoplast and in the vacuolar lumen. An identical pattern was observed for a truncated version of VSR1 lacking its intraluminal cargo-sorting domain (VSR1 (TMD-CT), Fig. S5c). Notably, almost all STX6A- and VSR1-positive structures were enlarged upon wortmannin treatment (Fig. 4e). These observations suggest that STX6A and VSR1 label wortmannin-sensitive TGN-MVB transitional compartments,

whereas VHA-A1 may function as a vacuole-specific V-ATPase. Consistent with this notion, VPS28A, VPS20, and VPS4A showed *c.* 20%, *c.* 50%, and *c.* 100% colocalization with STX6A or VSR1, respectively (Fig. 4f–j). In Arabidopsis, SYP61 predominantly localizes to the TGN, whereas VSRs mainly label MVBs (Sanderfoot *et al.*, 2001; Miao *et al.*, 2006; Scheuring *et al.*, 2011). The presence of moss STX6A/SYP61 on VSR1-positive MVBs in moss implies either that this reporter preferentially marks MVBs or it recognizes an MVB-associated subpopulation of TGN in a species-specific manner. Given that STX6/SYP61 is highly conserved and its *Marchantia* homolog

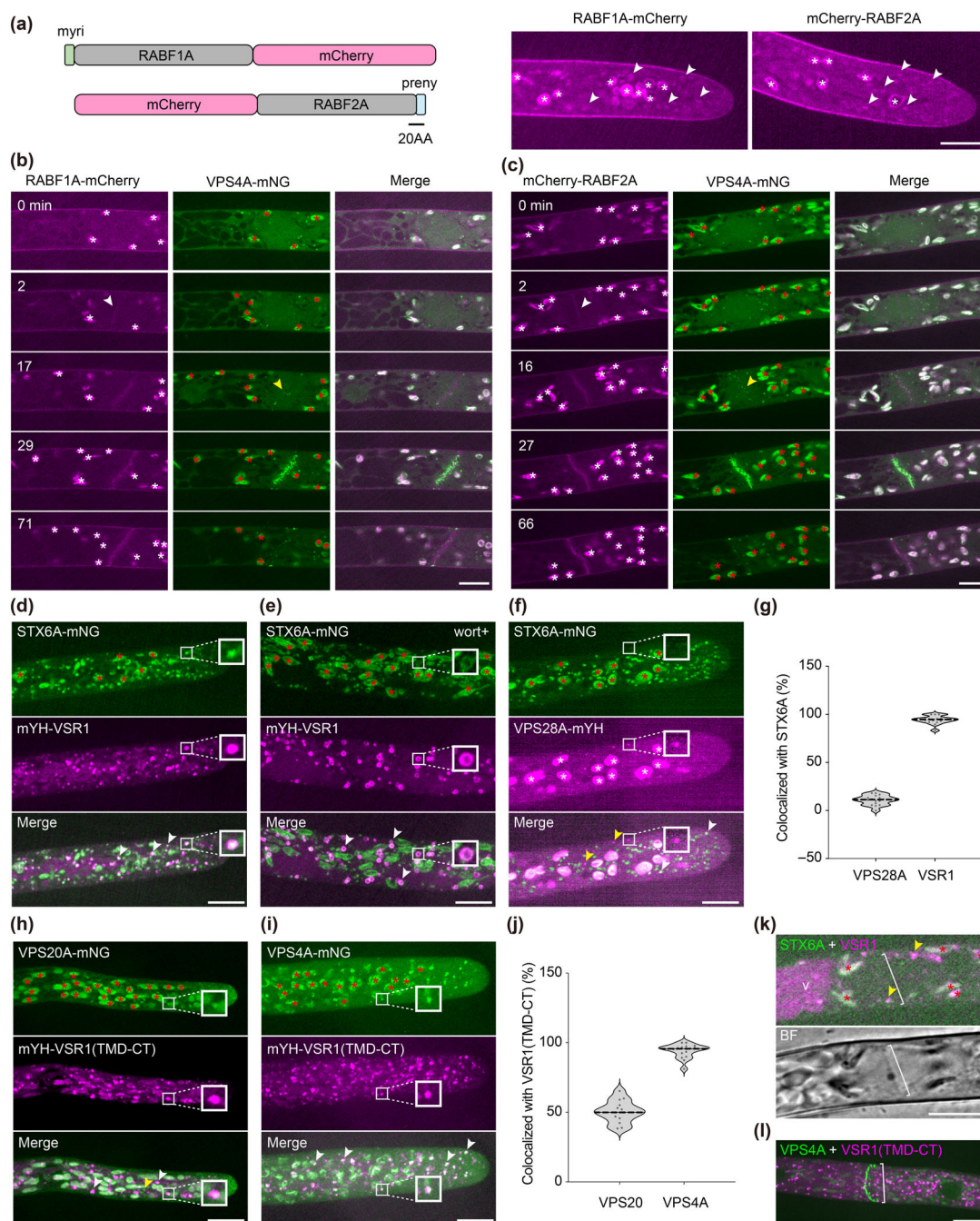


Fig. 4 The recruitment of VPS4A to the cell plate follows RAB5 GTPases but does not involve reporters for *trans*-Golgi network (TGN)-multivesicular body (MVB) transition. (a) The canonical (RABF2A) and plant-specific (RABF1A) RAB5 GTPases label the plasma membrane and vesicular structures (arrowheads) in *Physcomitrium patens* caulonemal tip cells. RABF1A contains an N-terminal myristoylation signal (myri), and RABF2A contains a C-terminal prenylation (preny) motif. To avoid membrane targeting, RABF1A was fused with a C-terminal mCherry tag, whereas RABF2A was fused with an N-terminal mCherry tag. Asterisks indicate chloroplast autofluorescence (a–f, h, i, k). (b, c) Localization of RABF1A (b) and RABF2A (c) at the cell plate. Note that RABF1A or RABF2A emerges at the midline of the cell plate (white arrowheads) earlier than VPS4A (yellow arrowheads). (d) Colocalization of the TGN marker STX6A with VSR1. Arrowheads indicate representative dual-labeled puncta. Enlarged views of a representative punctum are shown (d–f, h, i). (e) Wortmannin induces enlargement of vesicular compartments labeled by STX6A and VSR1 (arrowheads). (f) STX6A rarely colocalizes with VPS28A. Yellow arrowheads indicate VPS28A-positive puncta lacking STX6A signals; white arrowheads indicate colocalized puncta. (g) Quantification of VPS28A- or VSR1-positive structures colocalized with STX6A. VPS28A, $n = 16$; VSR1, $n = 15$. Violin plots show the data distribution, individual data points, minimum/maximum values, and the median (dashed line). (h) Partial colocalization between VPS20 and VSR1 (TMD-CT). The yellow arrowhead indicates a VPS20-positive punctum free of VSR1 signal; white arrowheads indicate colocalized puncta. (i) Strong colocalization of VPS4A with VSR1 (TMD-CT). (j) Quantification of VPS20- or VPS4A-positive structures colocalized with VSR1 (TMD-CT). VPS20, $n = 14$; VPS4A, $n = 16$. Violin plots show the data distribution, individual data points, minimum/maximum values, and the median (dashed line). (k) STX6A and VSR1 do not localize to the cell plate (brackets). Arrowheads indicate puncta labeled by STX6A-mNG and mYH-VSR1. v, vacuole. (l) VPS4A, but not VSR1 (TMD-CT), localizes to the cell plate (brackets). Bars, 10 μm (a–f, h, i, k, l).

localizes to the TGN (Kanazawa *et al.*, 2016), the latter scenario appears more consistent with the known function of this protein. Nevertheless, we analyzed the localization of STX6A, VSR1, and VHA-A1 during cell division and found that none of these reporters accumulated at the cell plate (Figs 4k,l, S5a,b,d). The absence of STX6A and VSR1 from the cell plate, despite the distinct localization of STX6A in moss, suggests that ESCRT function in cytokinesis may not involve local TGN-MVB transition at the division plane.

Given that RABF-positive MVBs and recycling endosomes have been both implicated in cell plate assembly in Arabidopsis (Dhonukshe *et al.*, 2006), we examined whether recycling endosomes play a similar role in *P. patens*. Unexpectedly, endogenous labeling of GNOM3, a moss homolog of the ARF-GEF required for endosome recycling (Geldner *et al.*, 2003; Yi & Goshima, 2020b), revealed no cell plate localization, although vesicle-like structures were observed in the cytosol (Fig. S6a,b). Treatment with 10 μ M brefeldin A (BFA), an inhibitor of ARF-GEFs (Nebenfuhr *et al.*, 2002), caused severe growth retardation and the formation of multiple growing tips (Fig. S6c). Under BFA treatment, RABF2A, but not RABF1A, accumulated in BFA body-like compartments. Despite these effects, the cell plate localization of VPS4A, RABF1A, or RABF2A was not affected, nor were any cytokinesis defects detected (Fig. S6c, $n = 45$ cells). These findings suggest that recycling endosomes are not required in cytokinesis in *P. patens*. Therefore, the cytokinetic functions of ESCRT-related proteins may not depend on a direct involvement of recycling endosomes as reported in Arabidopsis (Dhonukshe *et al.*, 2006).

VPS4 is essential for moss development and vacuolar trafficking

We next explored the function of ESCRT-related proteins by focusing on VPS4 and LIP5, given their pivotal roles in MVB biogenesis and membrane remodeling (Babst *et al.*, 1998; Nickerson *et al.*, 2006; Haas *et al.*, 2007; Lata *et al.*, 2008). To this end, we generated knockout mutants of VPS4 and LIP5 via homologous recombination. The moss genome encodes two VPS4 proteins (VPS4A and VPS4B) and a single LIP5. As in Arabidopsis, the *lip5* knockout mutant did not display observable developmental defects. No phenotypes were detected for *ups4a* or *ups4b* single mutants either. We were unable to generate a *ups4a ups4b* double mutant. Therefore, we expressed a dominant-negative form of VPS4A, carrying the glutamate-to-glutamine (E237Q) mutation (Babst *et al.*, 1997) in wild-type plants using a β -estradiol inducible system (Kubo *et al.*, 2013). The *ups4a* (E237Q) inducible expression lines (*ups4a*(E237Q)-*iox*) cultured on standard BCDAT medium developed normally but died when supplemented with 1 μ M β -estradiol (Fig. 5a,b), indicating an essential role of VPS4 in development.

We also generated a viable hypomorphic *ups4* mutant by introducing methionine-to-threonine (M312T) and leucine-to-serine (L332S) mutations into the endogenous *ups4a* site via genome editing in a *ups4b* knockout background (hereafter referred to as *ups4-hm*). These mutations render *ups4* partially loss-of-function

in yeast (Babst *et al.*, 1997), and likely impair hydrophobic contacts with neighboring residues in the small AAA domain, thereby reducing its structural stability (Fig. 5a). Accordingly, *ups4-hm* plants exhibited partially retarded growth (Fig. 5c–f). The plant growth phenotype was rescued by overexpressing VPS4B (Fig. 5f), demonstrating that VPS4A and VPS4B are functionally redundant.

We next examined vacuolar trafficking and vacuole homeostasis in *ups4* mutants. As shown in Fig. 5(g,h), vacuolar uptake of FM4-64 was delayed in *ups4-hm* and almost completely blocked in *ups4a*(E237Q)-*iox* lines. When *ups4a*(E237Q)-*iox* lines were transiently treated with 1 μ M β -estradiol, vacuole morphology in the tip region of caulonemal cells gradually changed from a tubular network into multiple enlarged, round-shaped compartments due to the fusion of vacuolar subdomains (Fig. 5i, Video S12). At the subcellular level, electron microscopy revealed a significant reduction in the number of ILVs in *ups4-hm* mutants (Fig. 5j,k), supporting the conserved role of VPS4 in ILV release (Haas *et al.*, 2007). Together, these findings indicate that VPS4 is essential for moss development and plays a conserved role in vacuolar trafficking.

VPS4 regulates cytokinesis progression

Because overexpression of VPS4A(E237Q) caused lethality, precluding direct analysis of VPS4's role in cell division, we examined the cell division process in *ups4-hm* mutants. In these mutants, mitosis and cytokinesis could still be completed, and the mitotic progression appeared unaffected. However, cytokinesis duration was significantly prolonged compared to wild-type cells (Fig. 6a,c; Videos S13, S14). This phenotype was rescued by overexpressing VPS4B fused with either an FLAG or an mNG tag (Fig. 6c), indicating that fluorescent protein tagging does not interfere with VPS4 function. We also developed a cell cycle-dependent inducible system to express the dominant-negative VPS4B(E236Q)-mNG fusion protein, wherein β -estradiol-binding chimeric transcription activator was expressed by the promoter of an A-type mitotic cyclin. Upon transient induction with 1 μ M β -estradiol, VPS4B(E236Q)-mNG was specifically expressed in dividing cells and localized to the cell plate (Fig. 6b). This led to a stronger delay in cytokinesis (Fig. 6c). These data together support that VPS4 is a key regulator of cytokinesis progression.

The removal of cell plate assembly factors was delayed in *ups4* mutants

We next investigated the dynamics of vesicular components required for cytokinesis, focusing on *P. patens* RABA1D and DRP1A. In flowering plants, RAB-A2 and RAB-A3 GTPases label secretory vesicles essential for cell plate assembly and are rapidly removed during cell plate expansion (Chow *et al.*, 2008). Dynamin-related proteins, particularly the DRP1 subfamily, regulate endocytosis and membrane tubulation (Gu & Verma, 1996, 1997; Otegui *et al.*, 2001; Kang *et al.*, 2003; Collings *et al.*, 2008; Boutte *et al.*, 2010). The *P. patens* genome encodes 13 RAB-A members and five DRP1 homologs. Although RAB-

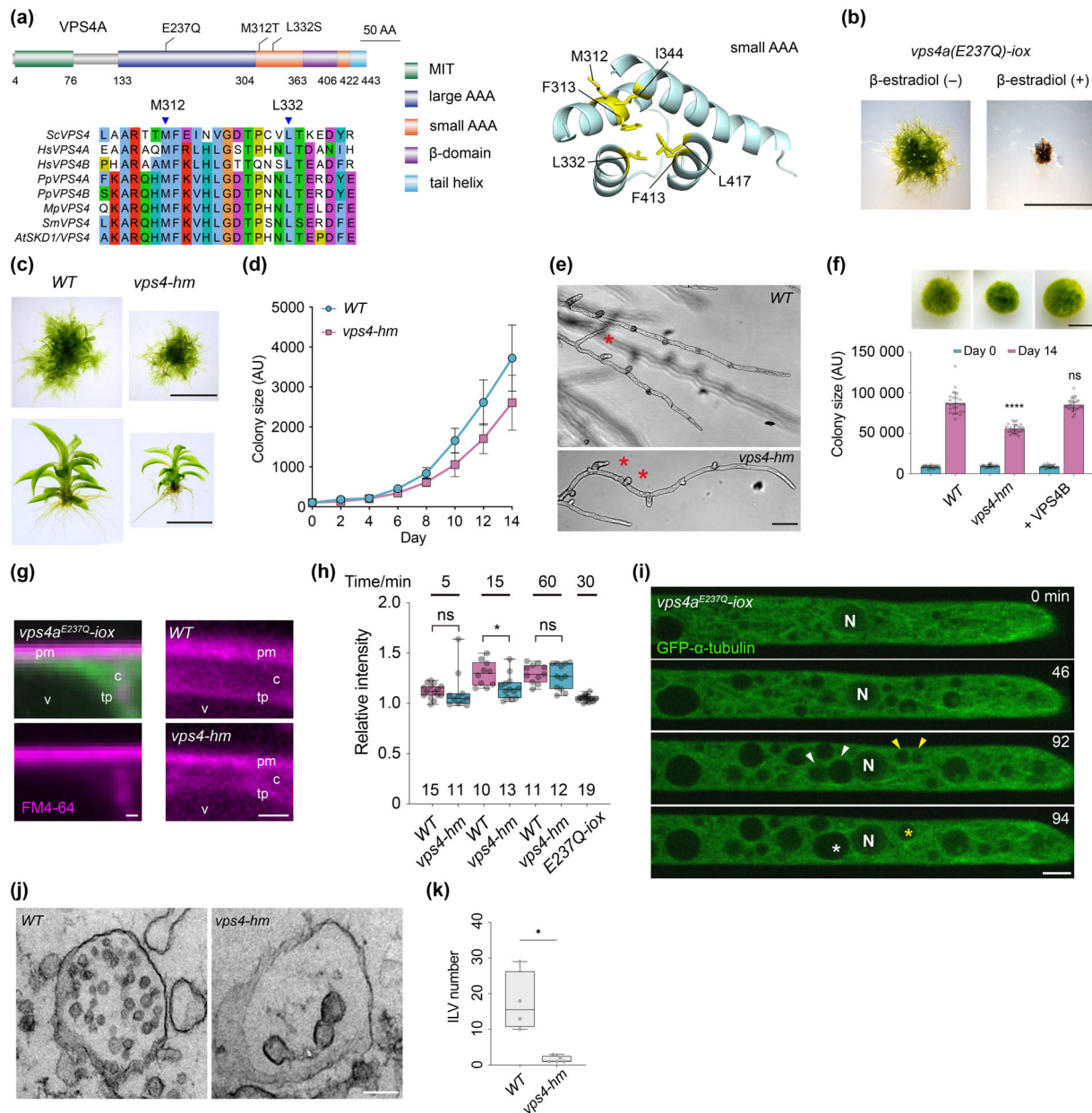


Fig. 5 VPS4 is essential for moss development and vacuolar trafficking. (a) Domain organization and key mutation sites of VPS4A. The M312 and L332 residues are highly conserved across species, including budding yeast *Saccharomyces cerevisiae* (ScVPS4), human *Homo sapiens* (HsVPS4A and HsVPS4B), moss *Physcomitrium patens* (PpVPS4A and PpVPS4B), liverwort *Marchantia polymorpha* (MpVPS4), lycophyte *Selaginella moellendorffii* (SmVPS4), and *Arabidopsis* (AtSKD1/VPS4). M312 and L332 are predicted to interact with hydrophobic residues inside the small AAA core domain. (b) Inducible expression of VPS4A(E237Q) causes lethality in *P. patens*. Bar, 0.5 cm. (c) Colony and gametophore development of wild-type (WT) and *vps4* hypomorphic mutant (*vps4-hm*) plants. Bars, 0.5 cm (upper), 0.2 cm (lower). (d) Quantification of colony size in WT and *vps4-hm* plants. Data are presented as mean $\pm$ SD. WT, $n = 48$; *vps4-hm*, $n = 32$. (e) Protonemal development in WT and *vps4-hm* mosses. Protonemal filaments in *vps4-hm* mutants are frequently wavy, and branch cells show retarded growth (asterisks). Bar, 100 μ m. (f) Quantification of colony size for WT, *vps4-hm* mutant, and rescue lines expressing VPS4B-FLAG. Data are presented as mean $\pm$ SD, with individual data points overlaid ($n = 25$ per genotype). At day 14, colony sizes were compared with WT using a Kruskal–Wallis test followed by Dunn's *post hoc* multiple comparisons; ns, not significant; ****, $P < 0.0001$. Bar, 0.5 cm. (g) Vacuolar trafficking of FM4-64 (magenta) is impaired in *vps4* mutants. The cytoplasm (green) in *vps4*^{E237Q-iox} is marked by GFP- α -tubulin. pm, plasma membrane; v, vacuole; tp, tonoplast; c, cytoplasm. Bars, 1 μ m. (h) Quantification of FM4-64 fluorescence intensity at the tonoplast, normalized to cytoplasmic signals. Cells were stained with FM4-64 for 5, 15, 30, and 60 min. Data are presented as box plots, showing the interquartile range (box), median (central line), and range (minimum–maximum; whiskers), with individual data points overlaid. Student's *t*-tests; ns, not significant; *, $P < 0.05$. The number of cells quantified is indicated below each bar. (i) Time-lapse images of caulonemal tip cells transiently treated with 1 μ M β -estradiol for 10 min. White and yellow arrowheads mark two fusion events; fused vacuolar compartments are indicated by asterisks. N, nucleus. Bar, 10 μ m. (j) Electron microscopy of multivesicular bodies (MVBs) in WT and *vps4-hm* mutant plants. Bar, 100 nm. (k) Quantification of ILV numbers. WT, $n = 4$ cells; *vps4-hm*, $n = 5$ cells. Data are presented as box plots, showing the interquartile range (box), median (central line), and range (minimum–maximum; whiskers), with individual data points overlaid. *, $P < 0.05$ (Mann–Whitney test).

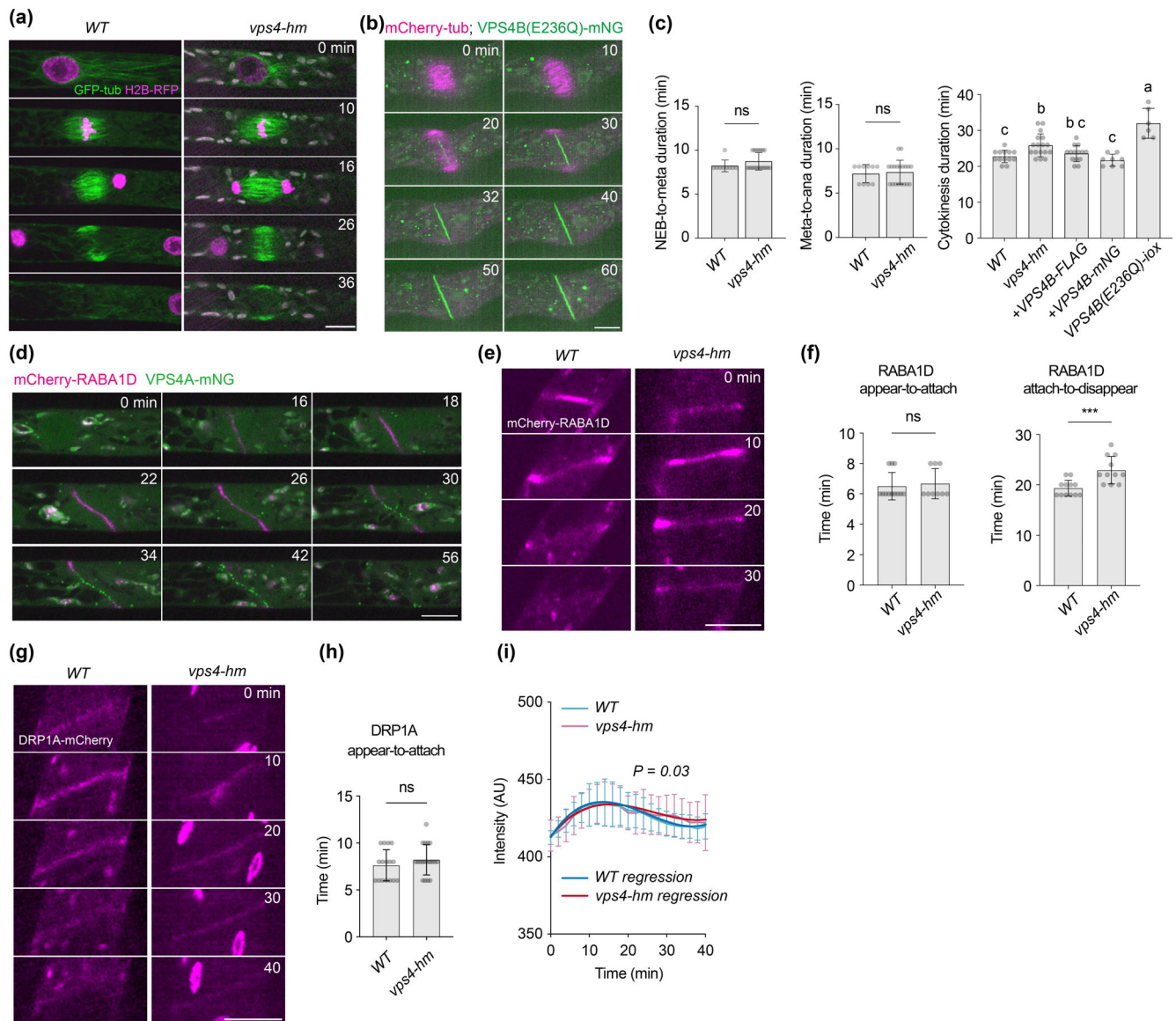


Fig. 6 VPS4 regulates cytokinesis progression. (a) Division of caulonemal tip cells in wild-type (WT) and *vps4-hm* *Physcomitrium patens* plants. Cells were labeled with GFP- α -tubulin and H2B-RFP. (b) Inducible expression of dominant-negative VPS4B(E236Q)-mNG in mitotic cells causes a strong delay in cytokinesis. (c) Quantification of the duration of cell division phases in WT, *vps4-hm* mutant, VPS4B(E236Q)-mNG inducible expression lines, and *vps4-hm* complemented with VPS4B-FLAG or VPS4B-mNG. Data are presented as mean $\pm$ SD, with individual data points overlaid. The durations of NEB-to-metaphase (NEB-to-meta) and metaphase-to-anaphase (meta-to-ana) were comparable between WT and *vps4-hm*. Mann–Whitney tests; ns, not significant; WT, $n = 9$; *vps4-hm*, $n = 19$. Cytokinesis duration was measured from the start of expansion to the disappearance of the phragmoplast. One-way ANOVA followed by Tukey's *post hoc* multiple comparisons; WT, $n = 13$; *vps4-hm*, $n = 17$; + VPS4B-FLAG, $n = 14$; + VPS4B-mNG, $n = 8$; VPS4B(E236Q)-iox, $n = 6$. (d) Localization of mCherry-RABA1D and VPS4A-mNG during cell division. (e) Localization of RABA1D in WT and *vps4-hm* mutant plants. (f) Quantification of the time from initial RABA1D localization to full cell plate expansion (appear-to-attach) and from attachment to the parental cell wall to complete disappearance (attach-to-disappear). Data are presented as mean $\pm$ SD, with individual data points overlaid. Appear-to-attach: WT, $n = 17$; *vps4-hm*, $n = 12$. Attach-to-disappear: WT, $n = 21$; *vps4-hm*, $n = 13$. Mann–Whitney tests; ns, not significant. ***, $P < 0.001$. (g) Localization of DRP1A in WT and *vps4-hm* mutant plants. (h) Quantification of the time from initial DRP1A localization to full cell plate expansion (appear-to-attach). Data are presented as mean $\pm$ SD, with individual data points overlaid. WT, $n = 16$; *vps4-hm*, $n = 11$. Mann–Whitney test; ns, not significant. (i) Quantification of DRP1A mean intensity at the cell plate over time. Data are presented as mean $\pm$ SD. A third-order polynomial regression was fitted to the data, and an extra sum-of-squares *F*-test revealed a significant difference between the curves ($P = 0.03$). WT, $n = 14$; *vps4-hm*, $n = 13$. Bars, 10 μ m (a, b, d, e, g).

A2 and -A3 both contribute to cytokinesis in Arabidopsis (Chow *et al.*, 2008), they fall into distinct subgroups when clustered with bryophyte RABs (Fig. S4). Phylogenetic analysis indicates that bryophyte RABs cluster into three major clades: RAB-A1, -A2,

and A3, which are related to AtRAB-A1/A2, AtRAB-A3/A4, and AtRAB-A5/A6, respectively (Fig. S4). RABA1D and DRP1A were selected for analysis based on their relatively high expression in caulonemal cells.

In wild-type cells, constitutively expressed mCherry-RABA1D was enriched at the leading edge of the cell plate and rapidly removed following cell plate expansion (Fig. 6d; Video S15),

mirroring the localization pattern of RAB-A2/A3 in *Arabidopsis* (Chow *et al.*, 2008). The removal of RABA1D occurred concurrently with the appearance of VPS4A (Fig. 6d) and was delayed

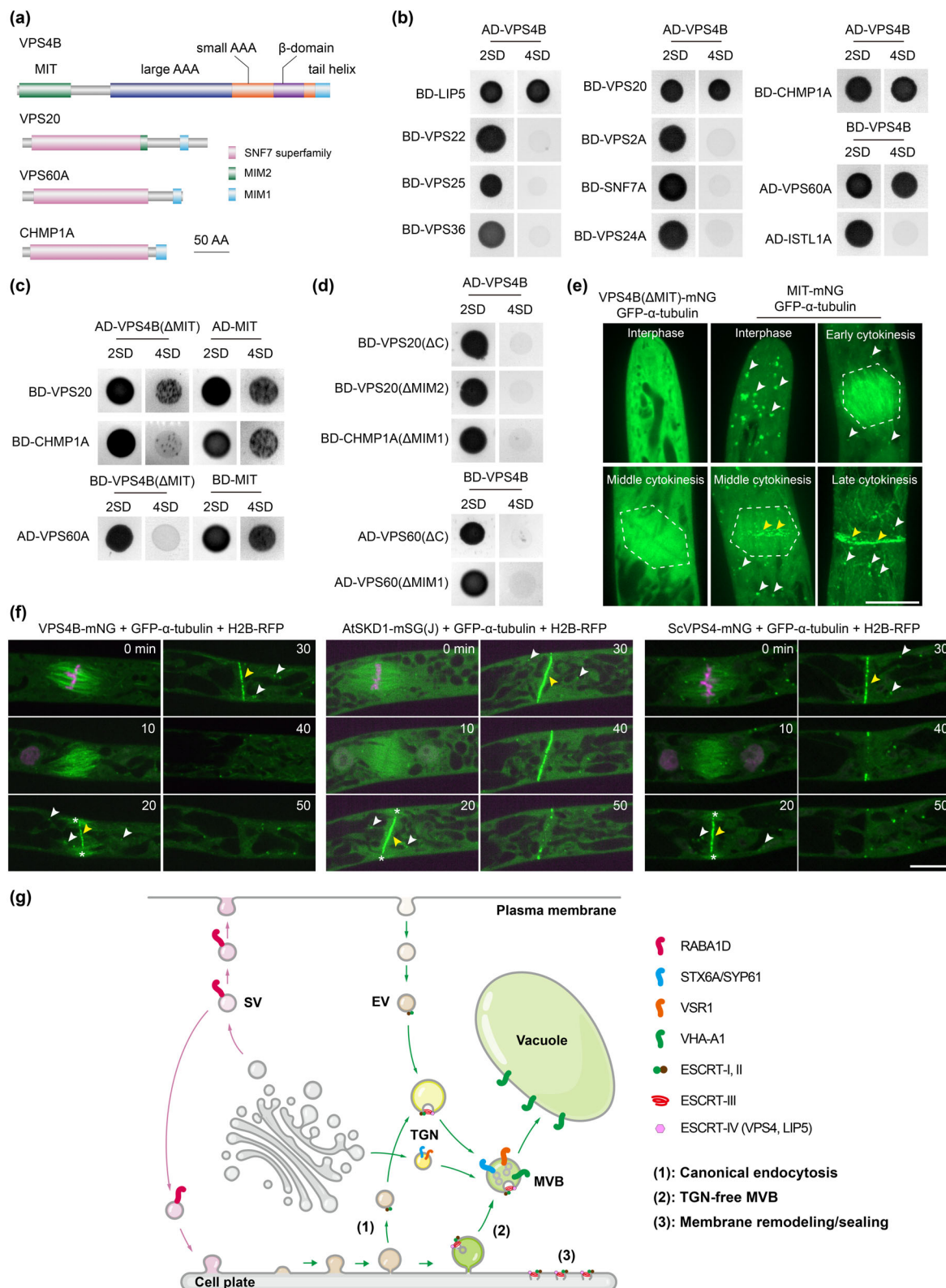


Fig. 7 Localization and function of VPS4 at the cell plate depend on canonical interactions with ESCRT-III components. (a) Domain organization of VPS4B, VPS20, VPS60A, and CHMP1A in *Physcomitrium patens*. VPS20, VPS60A, and CHMP1A contain an SNF7 superfamily domain and MIT-interacting motifs (MIMs). (b) Yeast two-hybrid assays showing that VPS4B interacts with LIP5, VPS20, VPS60A, and CHMP1A, but not with ESCRT-II subunits or other tested ESCRT-III subunits. 2SD: Trp-/Leu- synthetic dropout medium; 4SD: Trp-/Leu-/His-/Ade- synthetic dropout medium. (c) The MIT domain of VPS4B is necessary and sufficient for interactions with ESCRT-III subunits. (d) ESCRT-III subunits interact with VPS4B through their MIM motifs. (e) Localization of VPS4B(Δ MIT) and the MIT domain of VPS4B in moss. The indicated constructs were fused with a C-terminal mNG tag and constitutively expressed in the *vps4-hm* mutant to assess both functionality and localization. These lines also expressed a GFP- α -tubulin reporter to visualize microtubules and the phragmoplast. The tubulin-marked phragmoplast is outlined by a dashed polygon. Puncta labeled by VPS4B(Δ MIT) or the MIT domain are indicated by white (cytosolic) or yellow (at the cell plate) arrowheads. Bar, 10 μ m. (f) Localization of moss VPS4B (left), Arabidopsis AtSKD1/VPS4 (middle), and budding yeast ScVPS4 (right) during cell division. AtSKD1 is fused to the green fluorescent protein mStayGold(I) (mSG(I)). VPS4-positive puncta are denoted by white (cytosolic) or yellow (at the cell plate) arrowheads. Asterisks mark the leading edge of the phragmoplast. Bar, 10 μ m. (g) Proposed model for the localization and function of ESCRT-related proteins in moss. During interphase, ESCRT-I components are recruited to endocytic vesicles (EVs). These vesicles receive cargos from the TGN and mature into MVBs through the action of ESCRT-II, -III, and -IV (VPS4/LIP5). During cell division, ESCRT proteins may regulate cytokinesis via three mechanisms: (1) participation in canonical endocytosis; (2) generation of a TGN-independent population of MVBs; (3) direct involvement in membrane remodeling/sealing. ESCRT, Endosome Sorting Complexes Required for Transport.

in *vps4-hm* mutants (Fig. 6e,f; Videos S16, S17). Although DRP1A appeared later than RABA1D, its signal also gradually decreased during cytokinesis, consistent with the behavior of AtDRP1A (Fig. 6g; Video S18) (Kang *et al.*, 2003). Because DRP1A signal did not decrease as sharply as RABA1D did, we quantified its mean intensity over time and revealed a significant delay in DRP1A removal in the *vps4-hm* mutant (Fig. 6i; Video S19). The duration of RABA1D and DRP1A expansion from the cell center toward the parental wall were comparable between wild-type and *vps4-hm* lines (Fig. 6f,h). Similar results were obtained for another cell plate marker SCAMP4 (Fig. S7) (de Keijzer *et al.*, 2017). Moreover, the secretion of RABA1D-labeled vesicles was unaffected, as shown by fluorescence recovery after photobleaching assays (Fig. S8). Therefore, the defects in RABA1D and DRP1A removal may not result from delayed recruitment or impaired secretion. Together, our findings indicate that VPS4 and likely other ESCRT-related proteins associated with the cell plate are required for proper cytokinesis progression and regulate the dynamics of cell plate assembly factors.

Localization of VPS4A depends on canonical interactions with ESCRT-III

The involvement of all ESCRT complexes in cytokinesis suggests that VPS4 may be recruited through direct interactions with ESCRT-III subunits, as observed during MVB biogenesis in yeasts and animals (Babst *et al.*, 1998, 2002; Scott *et al.*, 2005; Nickerson *et al.*, 2006; Obita *et al.*, 2007; Stuchell-Brereton *et al.*, 2007; Adell *et al.*, 2014). The *P. patens* genome encodes a single copy of each ESCRT-II subunit and multiple copies of ESCRT-III core and accessory subunits, except for VPS20 and CHMP7 (Table S4). We selected representative proteins with relatively high expression in caulonemal cells (Fig. S1) and tested their ability to interact with VPS4B using yeast two-hybrid assays. Among the tested subunits, only VPS20, VPS60A, CHMP1A, and the VPS4 activator LIP5 showed binding with VPS4B (Fig. 7a,b). By contrast, no interactions were detected between VPS4B and ESCRT-II subunits, albeit ESCRT-II components interacted with ESCRT-I and ESCRT-III (Figs 7b, S9). In yeasts and animals, VPS4 binds ESCRT-III subunits primarily

through its microtubule-interacting and trafficking (MIT) domain, which recognizes MIT-interacting motifs (MIMs) present in ESCRT-III subunits (Babst *et al.*, 1998, 2002; Scott *et al.*, 2005; Nickerson *et al.*, 2006; Obita *et al.*, 2007; Stuchell-Brereton *et al.*, 2007; Adell *et al.*, 2014). Consistent with this notion, deletion of the MIT domain in VPS4B or of MIM motifs in ESCRT-III subunits significantly reduced their binding affinity (Fig. 7c,d). Furthermore, the MIT domain alone was sufficient to bind VPS20, VPS60A, and CHMP1A. In *P. patens*, VPS4B lacking the MIT domain completely lost its localization to MVBs and the cell plate (Fig. 7e). Conversely, the MIT domain alone was sufficient for correct localization to these compartments (Fig. 7e). Neither construct rescued the developmental defects of *vps4-hm* plants (Fig. S10). These findings underscore the conserved role of the MIT domain in VPS4 localization and function. Accordingly, Arabidopsis AtSKD1/VPS4 fused with mNG localized to the cell plate in moss similar to *P. patens* VPS4A (Fig. 7f; Video S20). Interestingly, ScVPS4 from budding yeast was targeted to the cell plate as well, even though yeast cell division involves membrane constriction and septum formation rather than inside-out cell plate expansion (Fig. 7f; Video S21). Together, these findings indicate that VPS4 function depends on conserved interactions with ESCRT-III during both MVB biogenesis and cytokinesis.

Discussion

In summary, we report that ESCRT-related proteins are conserved in the moss *P. patens* and play crucial roles in vacuolar trafficking. We also show that ESCRT proteins localize to the cell plate and provide evidence that VPS4 is essential for cytokinesis progression. Notably, factors involved in TGN-MVB transition preferentially associate with late MVBs and are not recruited to the division plane. Based on these findings, we propose that ESCRT-VPS4 may regulate cytokinesis independent of the canonical TGN-MVB-vacuole pathway (Fig. 7g).

In animals and archaea, ESCRT complexes drive membrane constriction to separate daughter cells, a function independent of their canonical roles in MVB biogenesis (Vietri *et al.*, 2020; Hurtig *et al.*, 2023). In Arabidopsis, functional loss of VPS23A or VPS4 causes multinucleate phenotypes, implying a critical

role for MVB regulators in cytokinesis (Spitzer *et al.*, 2006; Shahriari *et al.*, 2010). These observations together with delayed cytokinesis in moss *vps4* mutants suggest that ESCRT proteins and VPS4 may be widely involved in plant cell division. Because plant cell division does not involve membrane abscission, the underlying mechanism would differ from that in animals. Interestingly, ESCRT/VPS4 mutants in fission yeast also exhibit delayed daughter cell separation, implicating these factors in cytokinesis (Bhutta *et al.*, 2014). Fission yeast division relies on a unique constriction mechanism involving the assembly of a chitin-rich cell wall (septum), combining features of both animals (membrane constriction) and plants (wall assembly). These findings together point to an evolutionarily ancient role for MVB machinery – specifically ESCRT/VPS4 – in cell division, which may have been modified through lineage-specific adaptations. Future studies of these factors will provide valuable insights into the evolution of cytokinesis mechanisms across kingdoms.

If ESCRT machinery plays a fundamental role in plant cytokinesis, how does it regulate this process? There are several possibilities. First, ESCRT-positive MVBs may deliver endocytosed materials to facilitate cell plate assembly, as reported in Arabidopsis (Dhonukshe *et al.*, 2006). However, this model lacks strong support from multiple lines of evidence: the absence of MVB markers at the cell plate in Arabidopsis (Reichardt *et al.*, 2007; Chow *et al.*, 2008), the late recruitment of ESCRTs and VPS4 to the cell plate in *P. patens*, and the otherwise normal cell plate assembly observed in *vps4* mutants. Second, ESCRTs and VPS4 may mediate the degradation of cytokinetic materials, a possibility supported by evidence that the Qa-SNARE SYP111/KNOLLE is translocated to MVBs during late cytokinesis (Reichardt *et al.*, 2007; Chow *et al.*, 2008). In this model, MVBs could function either via the canonical endocytic route or through local maturation at the cell plate (Fig. 7g). Given that the TGN is absent from the cell plate in moss, the local maturation model would involve a unique, TGN-independent population of MVBs. In Arabidopsis, the localization of MVB biogenesis factors at the cell plate remains debated (Reichardt *et al.*, 2007; Chow *et al.*, 2008). Notably, TGN/EE markers such as VHA-A1 and SYP61 have been detected at the cell plate (Reichardt *et al.*, 2007; Steiner *et al.*, 2016). Therefore, MVBs may undergo local maturation to regulate cytokinesis in flowering plants, although this possibility requires functional validation.

Another potential role of ESCRT-VPS4 in cytokinesis is the regulation of membrane remodeling and sealing during cell plate maturation (Fig. 7g). In animals, ESCRT proteins participate in diverse membrane remodeling processes, including membrane repair, virus budding, and autophagosomal pore closure (Nakatogawa, 2020; Vietri *et al.*, 2020). In yeast, ESCRT-I to -III complexes, VPS4, and RAB5 GTPase are all required for autophagosomal pore closure (Nakatogawa, 2020). Given that plant cytokinesis involves extensive membrane remodeling, such as membrane fusion, tubulation, and likely hole closure (Sinclair *et al.*, 2023), a parallel mechanism for membrane remodeling and/or sealing is plausible. Supporting this idea,

ESCRT proteins, VPS4, and RAB5s are recruited to the cell plate following its expansion in *P. patens*, with some assembling into punctate structures potentially at sites of membrane remodeling and/or sealing (Fig. 3). In Arabidopsis, tubulation of fused vesicles during cell plate assembly is mediated by the dynamin GTPase DRP1 and its binding partner SH3P2, both of which can bend membranes (Otegui *et al.*, 2001; Kang *et al.*, 2003; Segui-Simarro *et al.*, 2004; Ahn *et al.*, 2017; Nagel *et al.*, 2017). Notably, SH3P2 also interacts with the ESCRT-I subunit VPS23A (Nagel *et al.*, 2017), providing a promising link between DRP1 and ESCRT proteins. The altered dynamics of DRP1A in *vps4* mutants implies a potential role for ESCRT-VPS4 in membrane remodeling following dynamin-mediated tubulation (Fig. 6g–i). However, since VPS23A does not localize to the cell plate in Arabidopsis (Nagel *et al.*, 2017), further comparative studies are needed to determine whether the DRP1-SH3P2-ESCRT pathway is conserved or differentially adapted between bryophytes and vascular plants. It should be noted that SH3P2 also possesses ubiquitin-binding ability (Nagel *et al.*, 2017); thus, it may additionally transfer ubiquitinated cargos to MVBs via ESCRT proteins for degradation. A dual function of ESCRT-VPS4 in membrane remodeling/sealing and material degradation is therefore possible. Given the opposing membrane topologies during endocytosis and membrane repair, future studies leveraging higher-resolution imaging techniques, such as super-resolution microscopy and immunogold electron microscopy, will be instrumental in further elucidating the underlying mechanisms.

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Competing interests

None declared.

Author contributions

PY conceptualized this study. JR, ZY, YG, LJ, CG and PY performed molecular biology. JR, ZY, YG and PY performed transgenesis. JR, ZY, YG and PY helped in imaging. JR, ZY, LJ and CG helped in yeast two-hybrid. JR helped in electron microscopy. JR and PY performed in data analysis. JR and PY contributed to manuscript writing. PY contributed to funding acquisition.

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Data availability

The data that support the findings of this study are available in the article, Figs S1–S10, Tables S1–S4, and Videos S1–S21. Sequence data used in this study are available from the Phytozome, TAIR, or UniProt databases with the following accession numbers: VPS4A, Pp6c1_790/Pp3c1_1310; VPS4B, Pp6c2_19890/Pp3c2_35940; LIP5, Pp6c12_360/Pp3c12_1420; VPS28A, Pp6c14_10940/Pp3c14_20060; VPS22, Pp6c10_7170/Pp3c10_13560; VPS25, Pp6c15_780/Pp3c15_1420; VPS36, Pp6c23_10660/Pp3c23_21630; VPS20, Pp6c2_17850/Pp3c2_32100; SNF7A, Pp6c3_19290/Pp3c3_36260; VPS24A, Pp6c9_9920/Pp3c9_19830; VPS2A, Pp6c23_7020/Pp3c23_14890; CHMP1A, Pp6c8_3480/Pp3c8_6470; VPS60A, Pp6c25_3260/Pp3c27_2180; ISTL1A, Pp6c2_7700/Pp3c2_13590; RABA1D, Pp6c1_14750/Pp3c1_28210; RABF1A, Pp3c11_7740/Pp6c11_4410; RABF2A, Pp3c6_28860/Pp6c6_15040; DRP1A, Pp3c22_1100/Pp6c22_560; GNOM3, Pp6c22_3190/Pp3c22_6150; STX6A, Pp6c17_12350/Pp3c17_23220; STX6B, Pp6c3_16380/Pp6c3_16380; VSR1, Pp6c11_1550/Pp3c11_2830; VHA-A1, Pp6c26_6100/Pp3c26_10750; SCAMP4, Pp6c1_11180/Pp3c1_21730; AtSKD1, AT2G27600; ScVPS4, YPR173C. A complete list of accession numbers for other ESCRT-related proteins in *P. patens* is available in Table S4.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Expression of ENDOSOMAL SORTING COMPLEX REQUIRED FOR TRANSPORT-related proteins.

Fig. S2 Effects of microtubule disruption on VPS4A localization.

Fig. S3 Localization of VPS4A during cell division.

Fig. S4 RAB GTPases in *Physcomitrium patens*.

Fig. S5 Localization of VHA-A1 and VSR1.

Fig. S6 Dispensability of recycling endosomes for cytokinesis.

Fig. S7 Cell-plate assembly in *vps4-hm* mutants.

Fig. S8 Secretory vesicles in *vps4-hm* mutants.

Fig. S9 Interactions between ENDOSOMAL SORTING COMPLEX REQUIRED FOR TRANSPORT subunits.

Fig. S10 Function of VPS4's MIT domain.

Table S1 Moss strains.

Table S2 Plasmids.

Table S3 Primers.

Table S4 ENDOSOMAL SORTING COMPLEX REQUIRED FOR TRANSPORT components.

Video S1 Localization of VPS28A-mNG.

Video S2 Localization of VPS36-mNG.

Video S3 Localization of VPS20-mNG.

Video S4 Localization of VPS4A-mNG.

Video S5 Localization of LIP5-mNG.

Video S6 Dynamics of VPS4A-mNG.

Video S7 Dynamics of LIP5-mNG.

Video S8 Dynamics of VPS4A-mNG and RABF1A-mCherry.

Video S9 Dynamics of VPS4A-mNG and mCherry-RABF2A.

Video S10 Localization of VPS4A-mNG and RABF1A-mCherry.

Video S11 Localization of VPS4A-mNG and mCherry-RABF2A.

Video S12 Vacuolar morphology in *vps4a*^{E237Q-iox}.

Video S13 Mitotic division in the *wild-type* plant.

Video S14 Mitotic division in the *vps4-hm* mutant.

Video S15 Localization of VPS4A-mNG and mCherry-RABA1D.

Video S16 Localization of mCherry-RABA1D in *wild-type* plants.

Video S17 Localization of mCherry-RABA1D in *vps4-hm* mutants.

Video S18 Localization of DRP1A-mCherry in *wild-type* plants.

Video S19 Localization of DRP1A-mCherry in *vps4-hm* mutants.

Video S20 Localization of AtSKD1-mNG.

Video S21 Localization of ScVPS4-mNG.

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New Phytologist Supporting Information

Article title: ENDOSOMAL SORTING COMPLEX REQUIRED FOR TRANSPORT machineries play crucial roles in vacuolar trafficking and cytokinesis in the moss *Physcomitrium patens*

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Article acceptance date: 14 September 2026

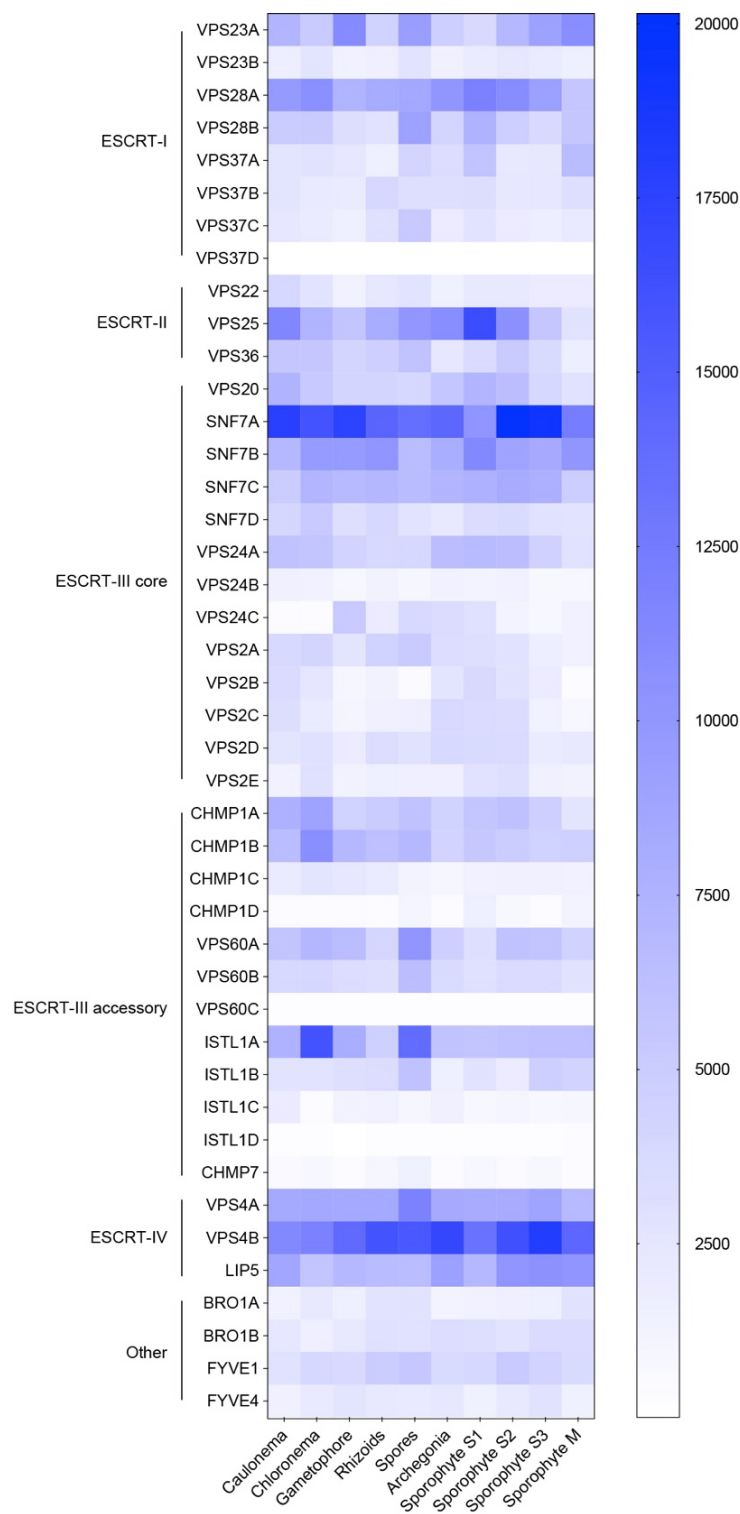


Fig. S1. Transcript expression levels of ESCRT-related proteins in the moss *P. patens*.

Expression data was extracted from the eFP browser database

(https://bar.utoronto.ca/efp_physcomitrella/cgi-bin/efpWeb.cgi).

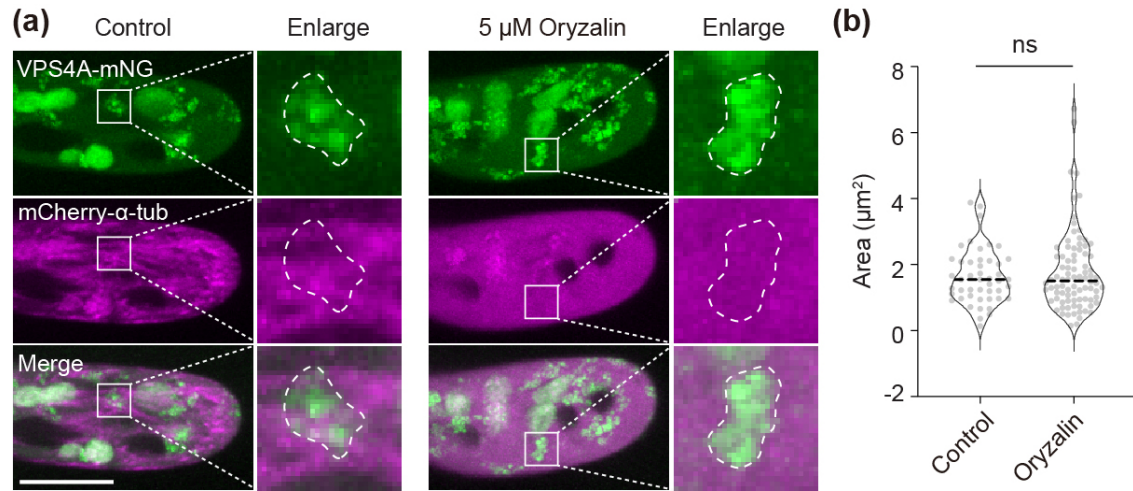


Fig. S2. Disruption of microtubule assembly with oryzalin does not impair the motility of VPS4A-labeled puncta.

(a) Maximum intensity projections from a 100-s movie of *Physcomitrium patens* caulonemal tip cells are shown. Representative regions are enlarged. Scale bar: 10 μ m.

(b) Quantification of the area occupied by VPS4A-labeled puncta. Violin plots show the data distribution, individual data points, minimum/maximum values, and the median (dashed line). Control: n=47; Oryzalin, n=86. Mann-Whitney test; ns, not significant.

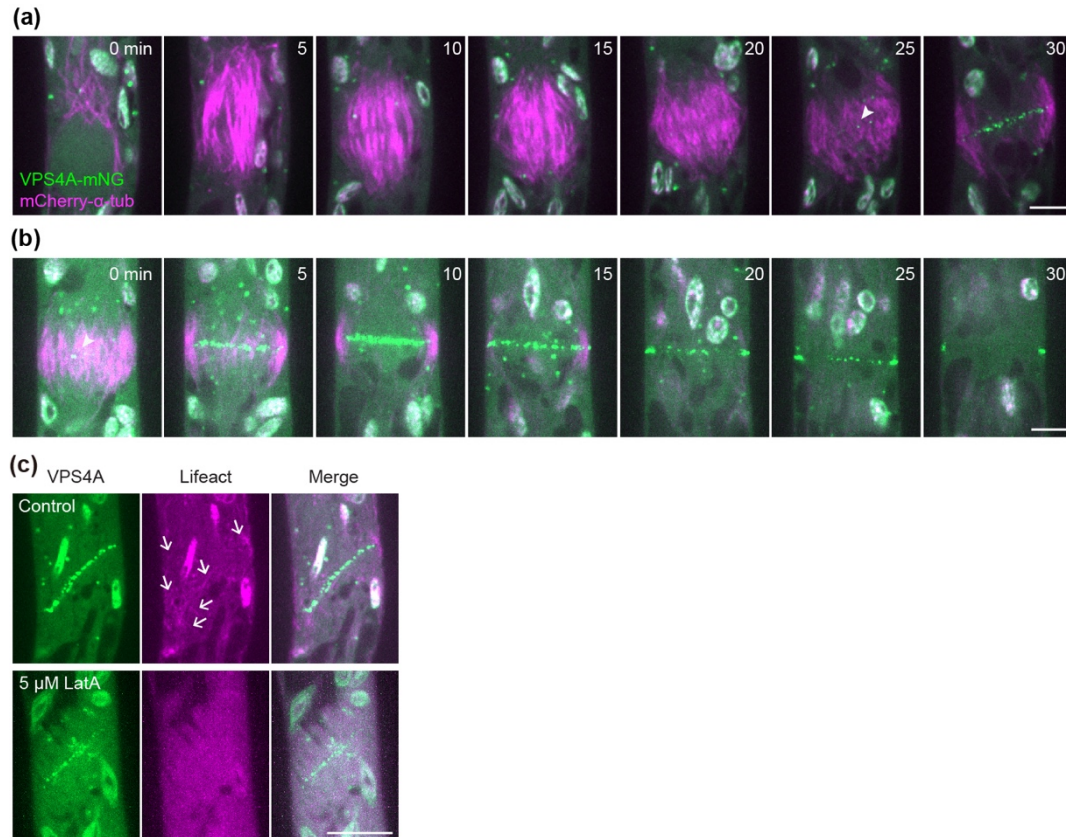
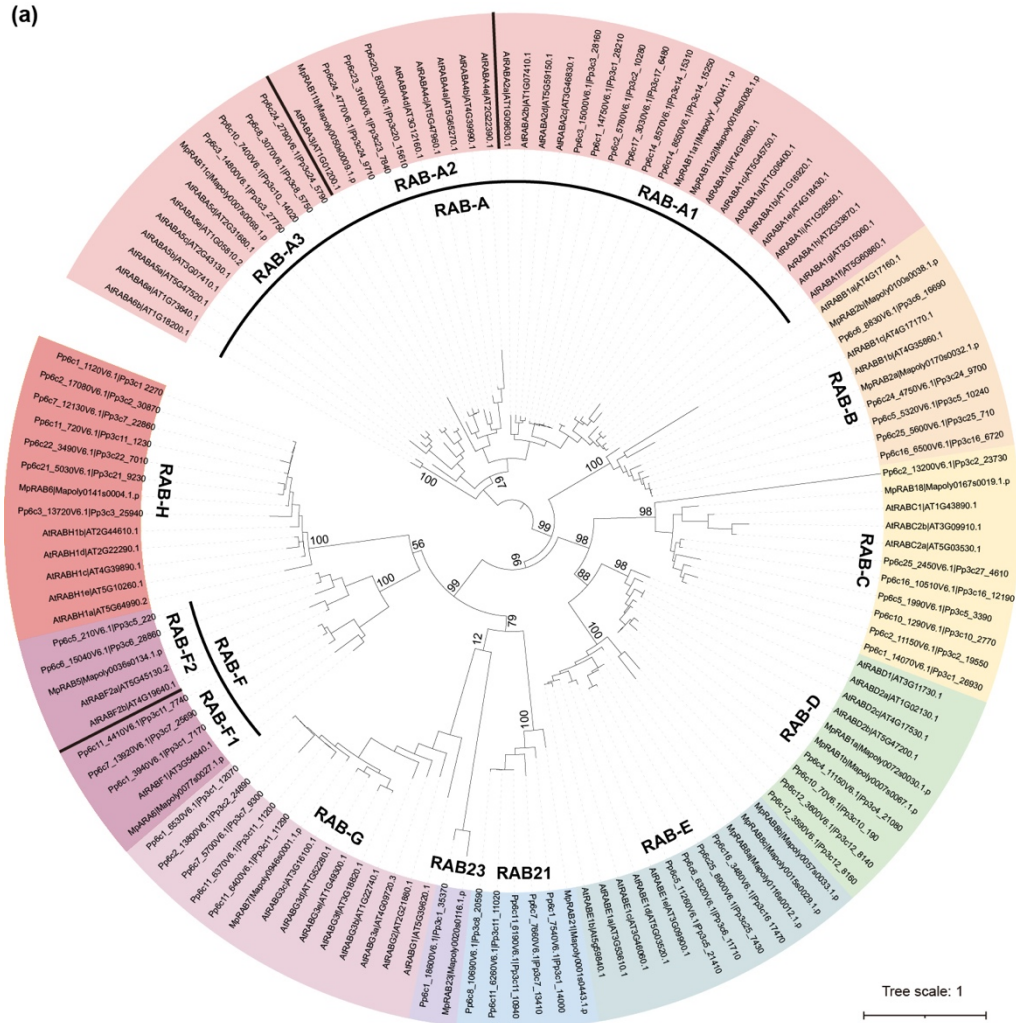


Fig. S3. Localization of VPS4A during cell division.

(a) Localization of VPS4A-mNG and microtubules (mCherry- α -tubulin) from nuclear envelope breakdown (NEB) through mid-cytokinesis in *Physcomitrium patens* caulonemal tip cells. The arrow indicates the initial appearance of VPS4A on the cell plate.

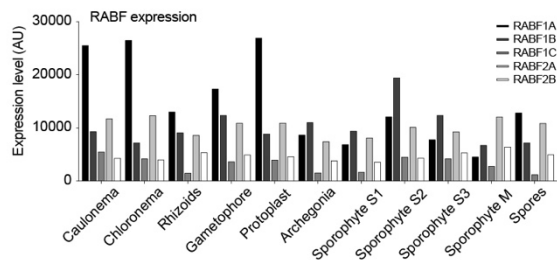
(b) Localization of VPS4A-mNG and mCherry- α -tubulin from early to late cytokinesis.

(c) Cell-plate localization of VPS4A was unaffected by latrunculin A (LatA) treatment. Arrows indicate actin filaments labeled with Lifeact-mCherry; these filaments were detectable in control cells but absent in LatA-treated cells. Scale bars: 10 μ m (a-c).



(b)

Gene	ID_v6.1	ID_v3.3	eFP transcript
RABF1A	Pp6c11_4410	Pp3c11_7740	Pp1s543_3V6.1
RABF1B	Pp6c1_3940	Pp3c1_7170	Pp1s38_328V6.1
RABF1C	Pp6c7_13920	Pp3c7_25690	Pp1s97_119V6.1
RABF2A	Pp6c6_15040	Pp3c6_28860	Pp1s117_91V6.1
RABF2B	Pp6c5_210	Pp3c5_220	Pp1s259_34V6.1



(c)

Gene	ID_v6.1	ID_v3.3	eFP transcript
RABA1A	Pp6c14_8550	Pp3c14_15250	Pp1s209_99V6.1
RABA1B	Pp6c14_8570	Pp3c14_15310	Pp1s209_94V6.1
RABA1C	Pp6c2_5780	Pp3c2_10280	Pp1s84_180V6.1
RABA1D	Pp6c1_14750	Pp3c1_28210	Pp1s21_298V6.1
RABA1E	Pp6c17_3030	Pp3c17_6480	Pp1s26_33V6.1
RABA1F	Pp6c3_15000	Pp3c3_28160	Pp1s157_53V6.1

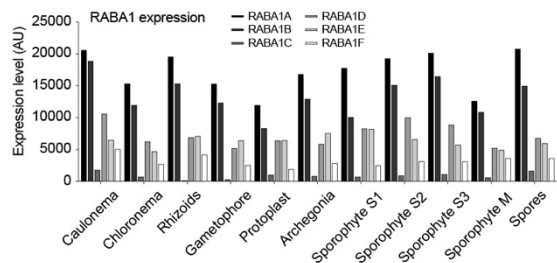


Fig. S4. Characterization of RAB GTPases in the moss *P. patens*.

(a) Maximum-likelihood phylogenetic tree of RAB GTPases from *P. patens*, the liverwort *M. polymorpha*, and *Arabidopsis*. Three major groups were identified in *P. patens* and *M. polymorpha*, corresponding to AtRAB-A1/A2, AtRAB-A3/A4, and AtRAB-A5/A6 in *Arabidopsis*, respectively. Because these groups could not be unambiguously divided into subgroups as in *Arabidopsis*, they were designated as RAB-A1, -A2, and -A3. Bootstrap values are indicated at major nodes.

(b-c) Gene IDs of RABF (b) and RABA1 (c) GTPases from different genome versions of *P. patens* and their expression levels in major tissues. Expression data was extracted from the eFP browser database (https://bar.utoronto.ca/efp_physcomitrella/cgi-bin/efpWeb.cgi).

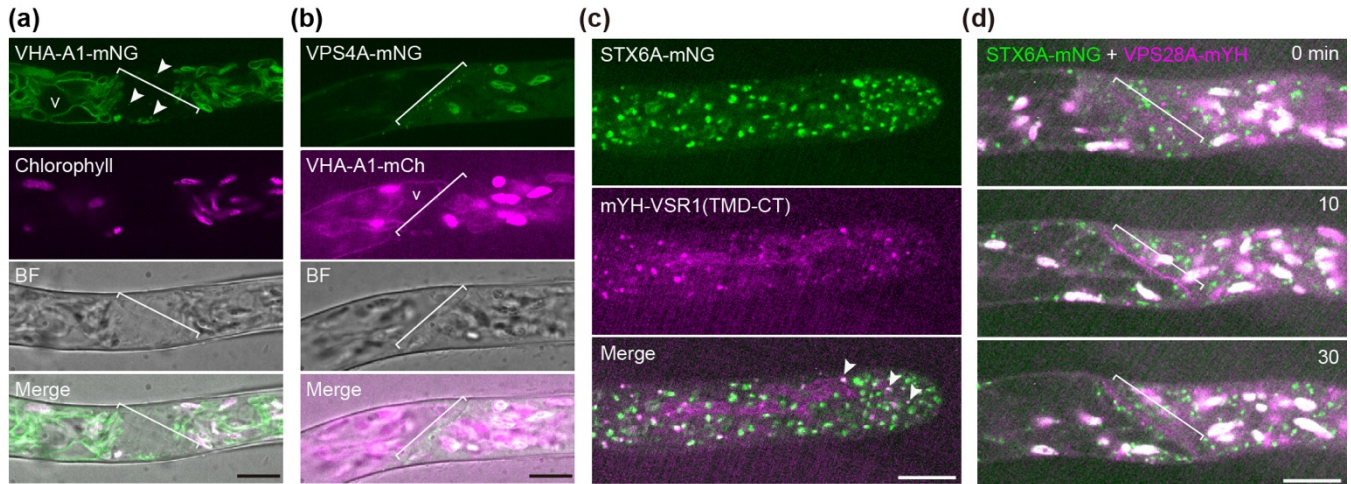


Fig. S5. VHA-A1 and VSR1 do not localize to the cell plate.

(a) VHA-A1 primarily localizes to the tonoplast and is absent from the cell plate (brackets) in *Physcomitrium patens* caulonemal tip cells. Arrowheads indicate punctate structures. BF, bright field; v, vacuole.

(b) Dual labeling with VPS4A-mNG and VHA-A1-mCherry confirms the absence of VHA-A1 at the cell plate (brackets).

(c) The truncated VSR1 protein, consisting of the transmembrane domain and C-terminal tail (TMD-CT), colocalizes with STX6A. Arrowheads indicate representative colocalized puncta.

(d) VPS28A, but not STX6A, localizes to the cell plate (brackets). Scale bars in (a-d): 10 μm.

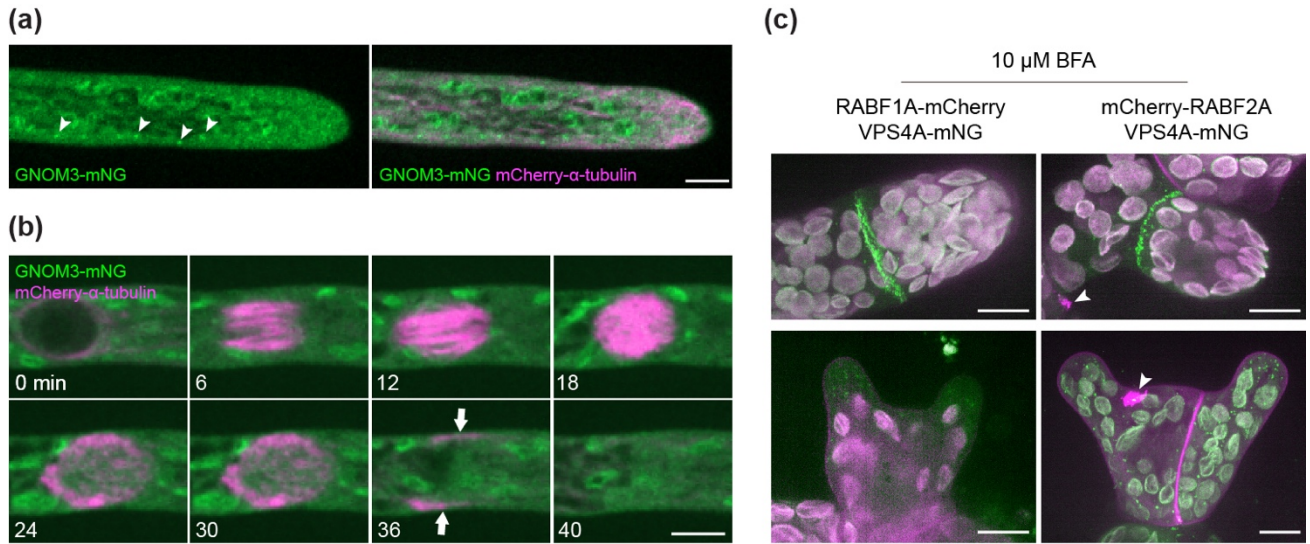


Fig. S6. Recycling endosomes do not regulate cytokinesis in *P. patens*.

(a) Localization of the ARF-GEF GNOM3 in *Physcomitrium patens* caulonemal tip cells. Arrowheads indicate vesicular structures.

(b) Localization of GNOM3 during cell division. Arrows indicate the cortical division zone.

(c) Localization of RABF1A-mCherry, mCherry-RABF2A, and VPS4A-mNG in protonemal cells cultured in the presence of 10 μM BFA. RABF2A was present in BFA body-like compartments (arrowheads). Scale bars: 10 μm (a-c).

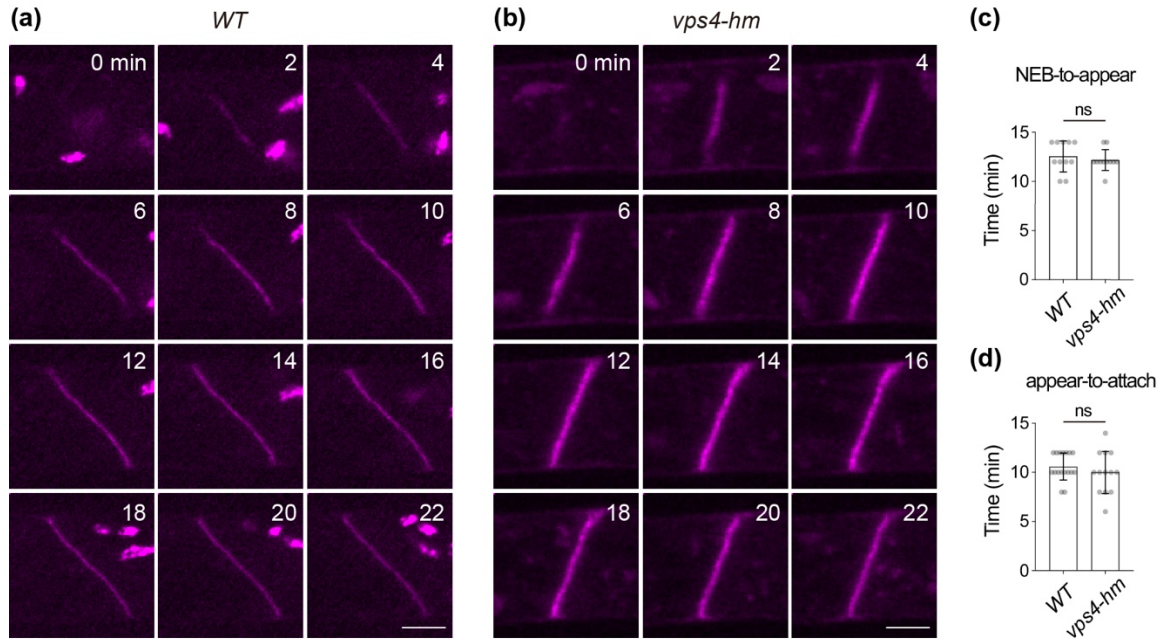


Fig. S7. Cell-plate assembly is unaffected in *vps4-hm* mutant plants.

(a-b) Cell-plate assembly in wild-type (WT) (a) and *vps4-hm* (b) *Physcomitrium patens* plants visualized with SCAMP4-mCherry. Scale bars: 10 μ m.

(c-d) Quantification of the time from nuclear envelope breakdown (NEB) to initial SCAMP4 localization at the cell center (NEB-to-appear) and from SCAMP4 initial appearance to full cell-plate expansion/attachment to the parental wall (appear-to-attach) in WT (c) and *vps4-hm* (d) plants. Sample sizes for NEB-to-appear: WT, n=11; *vps4-hm*, n=11. Sample sizes for appear-to-attach: WT, n=17; *vps4-hm*, n=13. Mann-Whitney tests; ns, not significant.

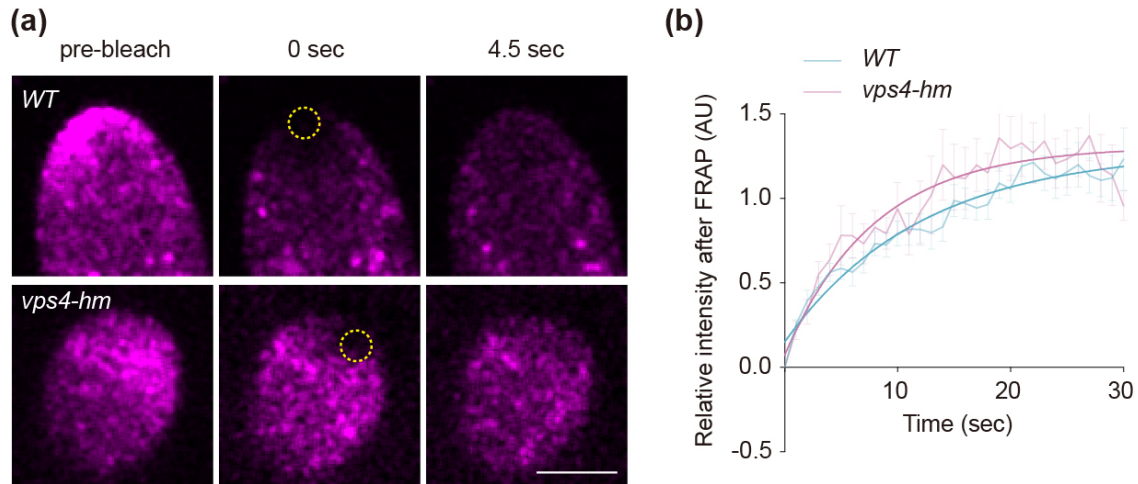


Fig. S8. Secretion of RABA1D-positive vesicles is not reduced in *vps4-hm* mutants.

(a) Fluorescence recovery after photobleaching (FRAP) assays of mCherry-RABA1D in *Physcomitrium patens* caulonemal tip cells. Photobleached regions are indicated by dashed circles. Scale bar: 5 μ m.

(b) Quantification of fluorescence intensity following photobleaching. Data are presented as mean $\pm$ SD. Intensity was normalized to signals at a neighboring non-bleached site and fitted with a one-phase association curve. Sample sizes: WT, n=14; *vps4-hm*, n=7.

Gene	VPS23A	VPS28A	VPS37A	VPS22	VPS25	VPS36	SNF7A	VPS20	VPS24A	VPS2A	CHMP1A	ISTL1A	VPS60A	VPS4B	LIP5
VPS23A	ND	+	+	-	-	-	ND	ND	ND	ND	ND	ND	ND	ND	ND
VPS28A		ND	+	-	-	-	ND	ND	ND	ND	ND	ND	ND	ND	ND
VPS37A			ND	+	-	-	-	-	-	-	-	-	-	-	ND
VPS22				ND	+	+	-	-	-	-	-	-	-	-	ND
VPS25					ND	+	-	+	-	-	-	-	-	-	ND
VPS36						ND	-	-	-	-	-	-	-	-	ND
SNF7A							ND	-	+	-	-	-	+	-	ND
VPS20								ND	-	-	-	-	-	+	ND
VPS24A									ND	-	-	-	+	-	ND
VPS2A										ND	-	-	-	-	ND
CHMP1A											ND	-	-	+	ND
ISTL1A												ND	-	-	ND
VPS60A													ND	+	ND
VPS4B														ND	+
LIP5															ND

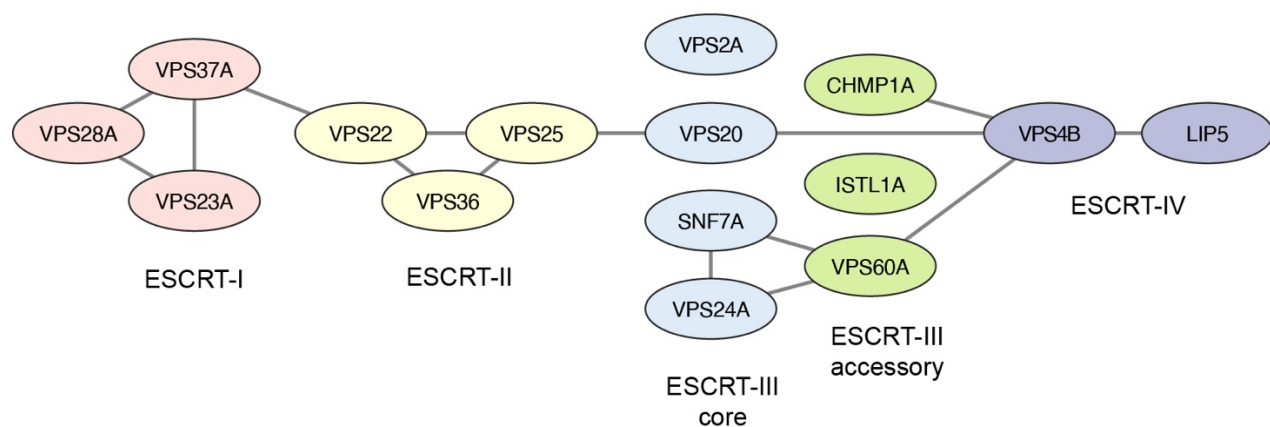


Fig. S9. Interactions between ESCRT subunits assessed by yeast two-hybrid assays.

Representative ESCRT components in *Physcomitrium patens* are investigated. VPS2A did not interact with any of the tested subunits, possibly due to weak binding affinity or preferential interaction with other subunits not included in this assay. +, positive interaction; -, no interaction; ND, not determined.

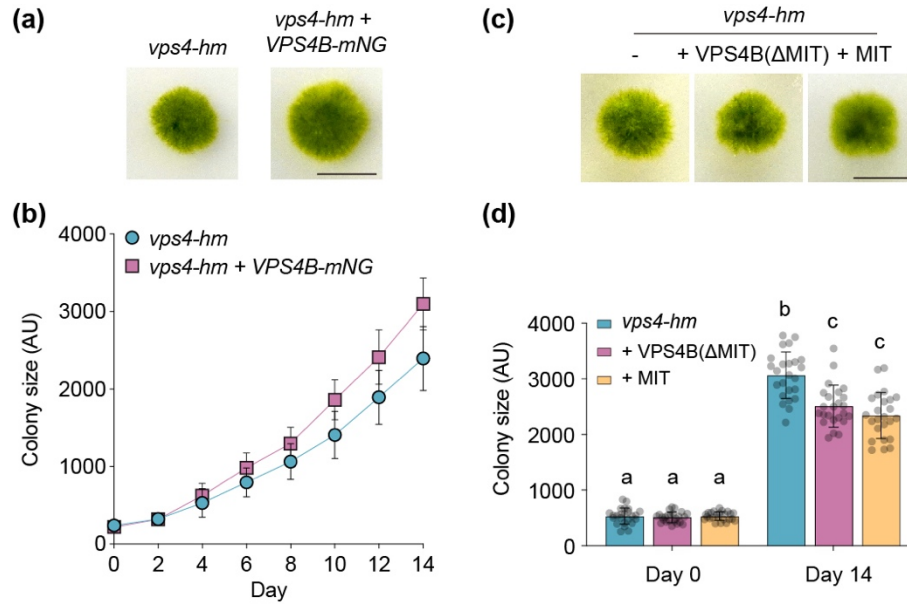


Fig. S10. The function of VPS4 requires its MIT domain.

(a) VPS4B-mNG rescues the colony growth defect in *vps4-hm* *Physcomitrium patens* plants.

(b) Quantification of colony size in *vps4-hm* and rescue lines. Data are presented as mean $\pm$ SD (n=23 per genotype).

(c) VPS4B(Δ MIT) and the MIT domain alone fail to rescue the colony growth phenotype of *vps4-hm*.

(d) Quantification of colony size in *vps4-hm* mutants and complemented lines. Data are presented as mean $\pm$ SD, with individual data points overlaid (n=24 per genotype). One-way ANOVA followed by Tukey's post-hoc multiple comparisons. Scale bars in (a) and (c): 0.5 cm.

Table S1. *Physcomitrium patens* strains used in this study.

Strain	Genotype
GH	HB7-EF1 α -GFP- α -tubulin;BS213-7113-HsH2B-mCherry
GPH0367	PTA1-EF1 α -mCherry- α -tubulin
GPH0407	PTA1-EF1 α -Lifeact-mCherry
GPH0738	GNOM3-mNG;PTA1-EF1 α -mCherry- α -tubulin
GPH0739	VPS4A-mNG;PTA1-EF1 α -Lifeact-mCherry
GPH0740	vps4a KO;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin
GPH0741	vps4b KO;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin
GPH0866	LIP5-mNG;PTA1-EF1 α -Lifeact-mCherry
GPH0921	PIG1-XVE-VPS4A(E237Q)-Cerulean;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin
PPH0003	vps4b KO;vps4a(M312T);BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin
PPH0004	vps4b KO;vps4a(L332S);BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin
PPH0009	vps4b KO;vps4a(M312T/L332S);BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin
PPH0053	VPS28A-mNG;PTA1-EF1 α -Lifeact-mCherry
PPH0054	VPS20-mNG
PPH0058	VPS20-mNG;PTA1-EF1 α -Lifeact-mCherry
PPH0066	VPS4A-mNG
PPH0070	VPS4A-mNG;PTA1-EF1 α -mCherry- α -tubulin
PPH0072	HB7-EF1 α -VPS4B-mCherry;VPS20-mNG
PPH0074	DRP1A-mCherry
PPH0082	PTA1-EF1 α -mCherry-RABA1D
PPH0086	PTA1-EF1 α -VPS4B-mNG;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0089	VPS4A-mNG;HB7-EF1 α -VPS20-mcherry
PPH0106	DRP1A-mCherry;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0107	PTA1-EF1 α -VPS4B(Δ MIT)-mNG;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0108	PTA1-EF1 α -VPS4B(MIT)-mNG;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0129	HB7-EF1 α -RABF1A-mCherry;VPS4A-mNG
PPH0130	HB7-EF1 α -mCherry-RABF2A;VPS4A-mNG
PPH0144	PTA1-EF1 α -mCherry-RABA1D;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0145	PTA1-EF1 α -ScVPS4-mNG;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0146	PTA1-EF1 α -AtSKD1-mSG(J);BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;vps4a(M312T/L332S);vps4b KO
PPH0149	HB7-EF1 α -VPS36-mNG
PPH0156	DRP1A-mCherry;VPS4A-mNG
PPH0157	SCAMP4-mCherry
PPH0166	HB7-EF1 α -mCherry-RABA1D;VPS4A-mNG

PPH0168	vps4a KO;BS213-7113-HsH2B-mCherry;HB7-EF1 α -GFP- α -tubulin;SCAMP4-mCherry
PPH0208	PIG1-XVE-CYCA2p-CYCA2N-VPS4B(E236Q)-mNG;PTA1-EF1 α -mCherry-tub
PPH0172	VHA-A1-mNG-Hyg
PPH0170	VHA-A1-mCherry-G418;VPS4A-mNG
PPH0248	HB7-EF1 α -STX6A-mNG-G418
PPH0268	PTA1-EF1 α -mYH-VSR1;HB7-EF1 α -STX6A-mNG
PPH0270	VPS28A-mYH;HB7-EF1 α -STX6A-mNG
PPH0368	VPS20-mNG;HB7-EF1 α -mYH-VSR1(TMD-CT)
PPH0369	VPS4A-mNG;HB7-EF1 α -mYH-VSR1(TMD-CT)

Table S2. Plasmids used in this study.

Plasmid	Description	Application
pPY54	VPS4A-KO	Knockout
pPY56	VPS4B-KO	Knockout
pPY154	LIP5-mNG	Knock-in
pPY208	VPS4A-sg1	CRISPR/Cas9-mediated mutation knock-in
pPY209	VPS4A-sg2	CRISPR/Cas9-mediated mutation knock-in
pPY311	VPS28A-mNG	Knock-in
pPY312	VPS20-mNG	Knock-in
pPY425	DRP1A-mCherry	Knock-in
pPY48	GNOM3-mNG	Knock-in
pPY55	VPS4A-mNG	Knock-in
pPY171	PIG1-XVE-VPS4A(E237Q)-Cerulean	Inducible expression
pPY419	PTA1-EF1 α -mCherry-RABA1D	Constitutive expression
pPY473	PTA1-EF1 α -VPS4B-mNG	Constitutive expression
pPY486	HB7-EF1 α -VPS4B-mcherry	Constitutive expression
pPY539	PTA1-EF1 α -VPS4B(Δ MIT)-mNG	Constitutive expression
pPY548	HB7-EF1 α -VPS20-mcherry	Constitutive expression
pPY584	HB7-EF1 α -RABF1a-mCherry	Constitutive expression
pPY594	HB7-EF1 α -mCherry-RABF2a	Constitutive expression
pPY638	PTA1-EF1 α -VPS4B(MIT)-mNG	Constitutive expression
pPY699	HB7-EF1 α -VPS36-mNG	Constitutive expression
pPY761	PTA1-EF1 α -ScVPS4-mNG	Constitutive expression
pPY780	PTA1-EF1 α -AtSKD1-mSG(J)	Constitutive expression
pPY328	AD-VPS20	Yeast two-hybrid
pPY329	AD-VPS25	Yeast two-hybrid
pPY331	AD-VPS28A	Yeast two-hybrid
pPY332	AD-LIP5	Yeast two-hybrid
pPY334	BD-VPS20	Yeast two-hybrid
pPY335	BD-VPS25	Yeast two-hybrid
pPY337	BD-VPS28A	Yeast two-hybrid
pPY338	BD-LIP5	Yeast two-hybrid
pPY346	AD-VPS4B	Yeast two-hybrid
pPY347	BD-VPS4B	Yeast two-hybrid
pPY443	AD-VPS2A	Yeast two-hybrid
pPY444	BD-VPS2A	Yeast two-hybrid
pPY446	AD-VPS24A	Yeast two-hybrid
pPY447	BD-VPS24A	Yeast two-hybrid
pPY450	BD-VPS60A	Yeast two-hybrid
pPY457	AD-VPS60A	Yeast two-hybrid
pPY461	BD-VPS22	Yeast two-hybrid
pPY463	AD-VPS36	Yeast two-hybrid

pPY464	AD-VPS22	Yeast two-hybrid
pPY466	BD-VPS36	Yeast two-hybrid
pPY577	BD-VPS4B(Δ MIT)	Yeast two-hybrid
pPY600	AD-VPS4B(Δ MIT)	Yeast two-hybrid
pPY601	AD-VPS4B(MIT)	Yeast two-hybrid
pPY602	AD-VPS60A(Δ MIM1)	Yeast two-hybrid
pPY603	BD-VPS4B(MIT)	Yeast two-hybrid
pPY604	BD-VPS20(Δ C)	Yeast two-hybrid
pPY605	AD-VPS60A(Δ C)	Yeast two-hybrid
pPY608	BD-VPS20(Δ MIM2)	Yeast two-hybrid
pPY598	HB7-EF1 α -mCherry-RABA1D	Constitutive expression
pPY530	DRP1A-mCherry	Knock-in
pPY939	SCAMP4-mCherry	Knock-in
pPY1141	PIG1-XVE-CYCA2p-CYCA2N-VPS4B(E236Q)-mNG	Inducible expression
pPY1105	VHA-A1-mNG-Hyg	Knock-in
pPY1104	VHA-A1-mCherry-G418	Knock-in
pPY1265	HB7-EF1 α -STX6A-mNG-G418	Constitutive expression
pPY1331	PTA1-EF1 α -mYH-VSR1	Constitutive expression
pPY1325	VPS28A-mYH	Knock-in
pPY1502	HB7-EF1 α -mYH-VSR1(TMD-CT)	Constitutive expression

Table S3. Primers used in this study.

Primer	Sequence(5'-3')	Application
oPY1 39	GATAAGCTTGATATCCTGATCATGTCATCAG GTGGTC	Amplify GNOM3 5' homology arm for C-terminal knock-in
oPY1 40	GCTTCCTCCTCCTCCTTGCGTTTCTTCAAAT GACTCATGCTC	Amplify GNOM3 5' homology arm for C-terminal knock-in
oPY1 41	CCCTCACACCGGTGAGGTTACATGTCTGTC CATGCAG	Amplify GNOM3 3' homology arm for C-terminal knock-in
oPY1 42	CATATGTCTAGACCCCAAGATCTGAGGAAG GAAGCTC	Amplify GNOM3 3' homology arm for C-terminal knock-in
oPY1 43	CTCTTGCCTTGTA CTTCAGTCAGG	Screen for GNOM3 knock-in lines
oPY1 44	TGGAGATGATTGACCCTCGATCTC	Screen for GNOM3 knock-in lines
oPY1 85	CCCCTCGAGGTCGACCATGTCCAAGTGATA ACATTGTGC	Amplify VPS4A 5' homology arm for knock out
oPY1 86	ATCGATACCGTCGACGGCTATCCTCTTCCC TCCAC	Amplify VPS4A 5' homology arm for knock out
oPY1 87	CCCTCACACCGGTGAAGCTGTTCCGATCTG CTAAGCC	Amplify VPS4A 3' homology arm for knock out and knock-in
oPY1 88	CATATGTCTAGACCCCATCGTAGTTCTGATG TTTTCTCCG	Amplify VPS4A 3' homology arm for knock out and knock-in
oPY1 89	CACTTTTGGGATTCTCTGCTGCTC	Screen for VPS4A knock out lines
oPY1 90	GATAATGTCCTGGACACAACGTG	Screen for VPS4A knock out and knock-in lines
oPY1 91	GATAAGCTTGATATCCCTATGTGTATACGTC AGTGCG	Amplify VPS4A 5' homology arm for knock-in
oPY1 92	ACTTCCACCTGATCCACCTTCTTCACCAAAC TCTTTGGT	Amplify VPS4A 5' homology arm for knock-in
oPY1 93	AGCTCCATGGATGCAGCGTAGT	Screen for VPS4A knock-in lines
oPY1 94	CCCCTCGAGGTCGACCGTGCAAGCTACAGA G TACTACAC	Amplify VPS4B 5' homology arm for knock out
oPY1 95	ATCGATACCGTCGACGGCTATCCCCTTCTC TACACACC	Amplify VPS4B 5' homology arm for knock out
oPY1 96	CCCTCACACCGGTGAGAAGGTTGAATGGTG CTTCGAG	Amplify VPS4B 3' homology arm for knock out
oPY1 97	CATATGTCTAGACCCCCAGAACTCACTTTCA CCTCCTG	Amplify VPS4B 3' homology arm for knock out
oPY1 98	GCATGTCTACCTATCCATTGCCATC	Screen for VPS4B knock out lines
oPY1 99	ACCCAATTGATGTGCGTGTCCG	Screen for VPS4B knock out lines
oPY4 17	CCCCTCGAGGTCGACGCTGACGGATTGTG GTATGAACAG	Amplify LIP5 5' homology arm for C-terminal knock-in
oPY4 18	ACTTCCACCTGATCCTATAGAGACAGAAGG AGATGTCAGC	Amplify LIP5 5' homology arm for C-terminal knock-in
oPY4 19	CCCTCACACCGGTGACTGCATGTTAATCTT GTGTATAGG	Amplify LIP5 3' homology arm for C-terminal knock-in

oPY4 20	CATATGTCTAGACCCCAGACCTTTCTCTATC TCGCCA	Amplify LIP5 3' homology arm for C-terminal knock-in
oPY4 21	CGTCCTACCATATCTAGCTTGTG	Screen for LIP5 knock-in lines
oPY4 60	GCCGCCCCCTTCACCATGTACAGCAACTTC AAAGAGCAG	Amplify VPS4A/B coding region for overexpression
oPY4 61	GCTTCCTCCTCCTCCACCTTCTTCACCAAAC TCTTTGGT	Amplify VPS4A coding region for overexpression
oPY8 27	GCTTCCTCCTCCTCCACCTTCTTCACCGAAC TCTTTGGT	Amplify VPS4B coding region for overexpression
oPY4 62	TCAATCTGATCTATGAAAATAATGG	Generate E237Q mutation in VPS4A
oPY4 63	CATAGATCAGATTGATTCCTTATG	Generate E237Q mutation in VPS4A
oPY1 059	CATAGACCAGATTGATTCTTTATGCGGT	Generate E236Q mutation in VPS4B
oPY1 060	TCAATCTGGTCTATGAAGATGATGGATG	Generate E236Q mutation in VPS4B
oPY4 64	GAGTAGGGTGTATTTGTAGCAGC	Sequence E237Q mutation
oPY4 78	CCCCTCGAGGTCGACGACTTGAAGGTGCCT TGTCAAAG	Amplify LIP5 5' homology arm for knock out
oPY4 79	GGGCTGCAGGAATTCGTTCTCGGTCATGC TCATCAT	Amplify LIP5 5' homology arm for knock out
oPY4 80	GCATGGCATTGTCCATTGTATTCAC	Screen for LIP5 knock out lines
oPY5 46	TCGGCAGCACATGTTTAGTTTTAGAGCTAGA AATAGCAAG	Generate sgRNA target 1 for introducing M312T mutation in VPS4A
oPY5 47	AACATGTGCTGCCGAGCCATGGGTTGTAAG TCCTCCAC	Generate sgRNA target 1 for introducing M312T mutation in VPS4A
oPY5 48	ATTATGAGGATCTGGCCGTTTTAGAGCTAG AAATAGCAAG	Generate sgRNA target 2 for introducing L332S mutation in VPS4A
oPY5 49	CCAGATCCTCATAATCGCATGGGTTGTAAG TCCTCCAC	Generate sgRNA target 2 for introducing L332S mutation in VPS4A
oPY5 50	GTTTAAGGCTCGGCAGCACACGTTTAAAGT GTCATGGTGTTC AATAT	Repair template for introducing M312T mutation in VPS4A
oPY5 51	CTGAACGCGATTATGAGGATAGTGCACGGA AGACTGATGGATTTTCA	Repair template for introducing L332S mutation in VPS4A
oPY6 96	CCCCTCGAGGTCGACTGTGATGGCTGAGTT CGAC	Amplify VPS20 5' homology arm for C-terminal knock-in
oPY6 97	GCTTCCTCCTCCTCCTGCAGCTAGTGGTTC TTCTA	Amplify VPS20 5' homology arm for C-terminal knock-in
oPY6 98	CCCTCACACCGGTGATTGCCTGTCAAGTGG AGCAT	Amplify VPS20 3' homology arm for C-terminal knock-in
oPY6 99	CATATGTCTAGACCCCCGAAC TACGACTTCT TGAC	Amplify VPS20 3' homology arm for C-terminal knock-in
oPY7 00	CAGAAGCTCATGGACGACAC	Screen for VPS20 knock-in lines
oPY7 01	GCTGGGTCAGTCAGCATTAG	Screen for VPS20 knock-in lines
oPY7 14	CCCCTCGAGGTCGACCACCTCCATGGATGC GTTGA	Amplify VPS28A 5' homology arm for C-terminal knock-in

oPY7 15	GCTTCCTCCTCCTCCAGAAGACTTGGGCAA TGA CTG	Amplify VPS28A 5' homology arm for C-terminal knock-in
oPY7 16	CCCTCACACCGGTGAGTTGCTTCCTGTACG ATGGTA	Amplify VPS28A 3' homology arm for C-terminal knock-in
oPY7 17	CATATGTCTAGACCCGATCACATCCTACATA GGGTC	Amplify VPS28A 3' homology arm for C-terminal knock-in
oPY7 18	ACGTACAAGATGGATTGCCC	Screen for VPS28A knock-in lines
oPY9 59	TAGGAGACTTCACCACAGATGT	Screen for VPS28A knock-in lines
oPY7 40	GGATCAGGAAGTGGTATGTACAGCAACTTC AAAGAGCAG	Amplify VPS4A/B coding region for overexpression
oPY7 41	GGATCAGGAAGTGGTATGGGTAATCTCTTG GTCAAGAAG	Amplify VPS20 coding region for overexpression
oPY7 44	GGATCAGGAAGTGGTATGGCGTATGCGGAA CAG	Amplify VPS28A coding region for overexpression
oPY7 45	GGATCAGGAAGTGGTATGATGAGCATGACC GAGGAAC	Amplify LIP5 coding region for overexpression
oPY7 46	GCTTCCTCCTCCTCCTATAGAGACAGAAGG AGATGTCAGC	Amplify LIP5 coding region for overexpression
oPY7 54	ATGTACAGCAACTTCAAAGAGC	Amplify VPS4A coding region
oPY7 55	ACCTTCTTCACCAA ACTCTTTG	Amplify VPS4A coding region
oPY1 079	GGATCAGGAAGTGGTGCTACA ACTGGCGCT AATCACACGCAAGC	Amplify RABF2A coding region for overexpression
oPY8 05	GCGCCACCCCTTTTAAGCGCAACATGTGGT TTTAGC	Amplify RABF2A coding region for overexpression
oPY8 06	GGATCAGGAAGTGGTGCGCATAAAGCGGAT GACGAG	Amplify RABA1D coding region for overexpression
oPY8 07	GCGCCACCCCTTTTAAGCCCCACAGCAGCC CTT	Amplify RABA1D coding region for overexpression
oPY8 25	GGAGGAGGAGGAAGCGCCACCACAGGCGC TAATCA	Amplify RABF2A coding region for overexpression
oPY8 27	GCTTCCTCCTCCTCCACCTTCTTCACCGAAC TCTTTGGT	Amplify VPS4B coding region for overexpression
oPY8 40	GAGGTCGACAAGCTTCAGAGGCAGTTGTTG ATGCA	Amplify DRP1A 5' homology arm for C-terminal knock-in
oPY8 41	GCTTCCTCCTCCTCCTTTCCAGGCCACAGC ATCAA	Amplify DRP1A 5' homology arm for C-terminal knock-in
oPY8 42	CCCTCACACCGGTGATCAACGGCTATACTC ATTTCTCC	Amplify DRP1A 3' homology arm for C-terminal knock-in
oPY8 43	CATATGTCTAGACCCAGTGTCTGTCTCTACA CGTC	Amplify DRP1A 3' homology arm for C-terminal knock-in
oPY8 46	CCCATCTTATAGCTCCTGAAC	Screen for DRP1A knock-in lines
oPY8 47	AGTGCAATGCTGGCCACTA	Screen for DRP1A knock-in lines
oPY8 49	GCCGCCCCCTTCACCATGGAGCAGCAGAA GCTGAGATCA	Remove MIT domain from VPS4B
oPY8 91	GGATCAGGAAGTGGTATGTTCAACCCCTTC ATCAG	Amplify VPS37A coding region for yeast-two-hybrid assay

oPY8 92	GCTTCCTCCTCCTCCCCAGGAGTAGTCGA GGA	Amplify VPS37A coding region for yeast-two-hybrid assay
oPY8 93	GGATCAGGAAGTGGTATGAGGAGGAGGCC AGGGATT	Amplify VPS22 coding region for yeast-two-hybrid assay
oPY8 94	GCTTCCTCCTCCTCCCATGTTGCCACCAGA TTCTGTCTC	Amplify VPS22 coding region for yeast-two-hybrid assay
oPY8 95	GGATCAGGAAGTGGTATGGAGGCCTGTGC AAGAGT	Amplify VPS36 coding region for yeast-two-hybrid assay
oPY8 96	GCTTCCTCCTCCTCCGCTGCCTTCTTGCAG GGTTGTAAT	Amplify VPS36 coding region for yeast-two-hybrid assay
oPY8 97	GGATCAGGAAGTGGTATGTTTTCTAGGATAT TCGGGAAGC	Amplify SNF7A coding region for yeast-two-hybrid assay
oPY8 98	GCTTCCTCCTCCTCCAAGAGCCATTTTCAGC TTGGAG	Amplify SNF7A coding region for yeast-two-hybrid assay
oPY8 99	GGATCAGGAAGTGGTATGGAGAAGGTGTTA AGCGT	Amplify VPS24A coding region for yeast-two-hybrid assay
oPY9 00	GCTTCCTCCTCCTCCTGAGCGCACACTGGC CAA	Amplify VPS24A coding region for yeast-two-hybrid assay
oPY9 01	GGATCAGGAAGTGGTATGAATATTTTAAAGA AGAAGCCGGAC	Amplify VPS2A coding region for yeast-two-hybrid assay
oPY9 02	GCTTCCTCCTCCTCCGCCACTTCGTAAAGC AGCT	Amplify VPS2A coding region for yeast-two-hybrid assay
oPY9 03	GGATCAGGAAGTGGTATGGGGGGCAGCCA GGAGAAGCTT	Amplify CHMP1A coding region for yeast-two-hybrid assay
oPY9 04	GCTTCCTCCTCCTCCACTCCTGTTTTTAAAGC TCGGCTAA	Amplify CHMP1A coding region for yeast-two-hybrid assay
oPY9 05	GGATCAGGAAGTGGTATGAGGCGCGTCTTC GGT	Amplify VPS60A coding region for yeast-two-hybrid assay
oPY9 06	GCTTCCTCCTCCTCCCGTTTCGTATCCCTGC CCTT	Amplify VPS60A coding region for yeast-two-hybrid assay
oPY9 07	GGATCAGGAAGTGGTATGTGGGAATCGATG CTTGGAAG	Amplify ISTL1A coding region for yeast-two-hybrid assay
oPY9 08	GCTTCCTCCTCCTCCCCGCCGCTTTGCCTG TTTCAGA	Amplify ISTL1A coding region for yeast-two-hybrid assay
oPY1 091	GCTTCCTCCTCCTCCGACCTGGACCTTCTT CTTGACCAA	Amplify VPS20 N-terminal membrane targeting region
oPY1 092	GGATCAGGAAGTGGTGAGCAGCAGAAGCT GAGATCAG	Remove MIT domain from VPS4B
oPY1 093	GCCGCCCCCTTCACCATGGGCTGTTTCATCA TCTTCTC	Amplify RABF1A coding region for overexpression
oPY1 094	GCTTCCTCCTCCTCCGGCCCGAATTGAGGG ATGGA	Amplify RABF1A coding region for overexpression
oPY1 098	ATAGTCGTTGTAGCCACTAGTC	Amplify RABF1A cDNA
oPY1 099	GTTCAATTCCTTGACAAGCTTG	Amplify RABF1A cDNA
oPY1 106	GGTGTCAACCAGGAGCTTTG	Amplify RABF2A cDNA
oPY1 107	CCTGCTAACAGCACCTGAC	Amplify RABF2A cDNA
oPY1 115	GCTTCCTCCTCCTCCATCCTCGCTGTCACC ATCACC	Amplify MIT domain of VPS4B

oPY1 116	GCTTCCTCCTCCTCCTGCCATTAGTGTCTCA AGTTCGTCG	Remove C-terminal region of VPS20
oPY1 117	GCTTCCTCCTCCTCCATCATTAGCTTCATGA TCTTGAGAATACTC	Remove C-terminal MIM1 motif of VPS20
oPY1 118	GCTTCCTCCTCCTCCAGATACAGTTGATTCC GCATCGC	Remove C-terminal MIM1 motif of CHMP1A
oPY1 119	GCTTCCTCCTCCTCCCATGTCGGCTTCCAA AGCATCC	Remove C-terminal region of VPS60A
oPY1 120	GCTTCCTCCTCCTCCTTCTGCAGCCTGCGG CTGAG	Remove C-terminal MIM1 motif of VPS60A
oPY1 179	ACCACTTCCTGATCCTGCAGCTAGTGGTTC TTCTA	Amplify VPS20 coding region for yeast-two-hybrid assay
oPY1 293	CGATCTTCGTCTCCAATTCG	Amplify AtSKD1 cDNA
oPY1 294	GATGCTTGATGGGAATGTCAAATC	Amplify AtSKD1 cDNA
oPY1 295	GCCGCCCCCTTCACCATGTACAGCAATTC AAGGAACAAGC	Amplify AtSKD1 coding region for overexpression
oPY1 296	CCACTTCCTGATCCACCTTCTTCTCCAACT CCTGTG	Amplify AtSKD1 coding region for overexpression
oPY1 285	GGATCAGGAAGTGGTATGAGCACGGGAGA TTTTTTAACTAAGG	Amplify ScVPS4 coding region for overexpression
oPY1 286	GCTTCCTCCTCCTCCGTTACCTTCTTGACCA AAATCTCTAGTG	Amplify ScVPS4 coding region for overexpression
oPY1 510	GAGGTCGACAAGCTTGGACCTCATGACAAG ATTCGAG	Amplify SCAMP4 5' homology arm for C-terminal knock-in
oPY1 511	GCTTCCTCCTCCTCCGATAGCTACTCTGCTT GCTGCA	Amplify SCAMP4 5' homology arm for C-terminal knock-in
oPY1 512	CCCTCACACCGGTGATTTTACTGTATTCTTA TTGTTAGTTGAGG	Amplify SCAMP4 3' homology arm for C-terminal knock-in
oPY1 513	CATATGTCTAGACCCCATCATACCACTTCTC AAACACATG	Amplify SCAMP4 3' homology arm for C-terminal knock-in
oPY1 514	TGGTAATTGAACGTCGGTGC	Screen for SCAMP4 knock-in lines
oPY1 515	GTGATCAACGTGGGATTAAAGAAGTCAC	Screen for SCAMP4 knock-in lines
oPY7 85	CTGTTACGACCATGCTTCG	Replace KINDa promoter in pPGX8
oPY1 691	GGAGGAGGAGGAAGCATGAAAGCGTTAAC GGCCAGG	Replace KINDa promoter in pPGX8
oPY1 692	CATGGTCGTGAACAGCTACAGAGAGCAGCA TACAGATG	Amplify CYCA2 promoter and N-terminal domain
oPY1 441	GCTTCCTCCTCCTCCAGAATGGCCCGTGTC TAAGTC	Amplify CYCA2 promoter and N-terminal domain
oPY1 753	CCCCTCGAGGTCGACTGTTGCGAAGGCAG CATATGCAG	Amplify VHA-A1 5' homology arm for C-terminal knock-in
oPY1 754	GCTTCCTCCTCCTCCCGCATCATCTTCTCT GAGCAAGTG	Amplify VHA-A1 5' homology arm for C-terminal knock-in
oPY1 755	CCCTCACACCGGTGATGGACATCTTGCAGA GGATTGGA	Amplify VHA-A1 3' homology arm for C-terminal knock-in
oPY1 756	CATATGTCTAGACCCCGTCGTGGATTATCG TCATAGCTG	Amplify VHA-A1 3' homology arm for C-terminal knock-in

oPY1 757	GAAGATAACCAGCTCTTCTCTGGC	Screen for VHA-A1 knock-in lines
oPY1 758	TGGCTTCGTGATTGCCTTGC	Screen for VHA-A1 knock-in lines
oPY2 124	GAAGGGTGTGAATAGAGCAG	Amplify VSR1 cDNA
oPY2 125	TCGCCACAAGTCAACCATC	Amplify VSR1 cDNA
oPY2 126	GCCGCCCCCTTCACCATGTGGACCATGGGA CGGT	Amplify VSR1 coding sequence for overexpression
oPY2 127	GCTTCCTCCTCCTCCAAACGTACTCCACGC GGGCAC	Amplify VSR1 signal peptide for overexpression
oPY2 128	GGATCAGGAAGTGGTGTGTGGAGGAGTCA GCTTTGATG	Amplify VSR1 coding sequence downstream of signal peptide for overexpression
oPY2 129	GCGCCCACCCTTTTATATCTCGTCCGCCAT GTGGA	Amplify VSR1 coding sequence for overexpression
oPY2 130	GGATCAGGAAGTGGTAAGCAGAGCCAGCC ATCGAAAC	Amplify VSR1 transmembrane domain for overexpression
oPY1 996	GCCGCCCCCTTCACCATGTCGGCGTTAGAT CCCTTC	Amplify STX6A coding region for overexpression
oPY1 851	GCTTCCTCCTCCTCCACCTGAAAATACAAGA AGAGTAAGAATTAGC	Amplify STX6A coding region for overexpression

Table S4. ESCRT components in representative eukaryotic species.

Regulators^a	Yeast (Sc)	Human (Hs)	Arabidopsis (At)	Physcomitrella (Pp)^b	eFP ID
ESCRT-0	VPS27	HGS	-	-	
	HSE1	STAM1 STAM2	-	-	
ESCRT-I	VPS23/STP22	TSG101	VPS23A/ELC (AT3G12400) VPS23B (AT5G13860)	VPS23A (Pp3c27_3040 / Pp6c25_3960) VPS23B (Pp3c16_2240 / Pp6c16_8110)	Pp1s54_243V 6.1 Pp1s242_76V 6.1
	VPS28	hVPS28	VPS28-1 (AT4G21560) VPS28-2 (AT4G05000)	VPS28A (Pp3c14_20060 / Pp6c14_10940) VPS28B (Pp3c17_18750 / Pp6c17_9710)	Pp1s34_161V 6.1 Pp1s65_107V 6.1
	VPS37/SRN2	VPS37A VPS37B VPS37C VPS37D	VPS37-1 (AT3G53120) VPS37-2 (AT2G36680)	VPS37A (Pp3c2_570 / Pp6c2_350) VPS37B (Pp3c1_40630 / Pp6c1_21370) VPS37C (Pp3c14_24180 / Pp6c14_13390) VPS37D (Pp3c17_23280 / Pp6c17_12380)	Pp1s226_46V 6.1 Pp1s424_28V 6.1 Pp1s69_112V 6.1 Pp3c17_2328 0V3.1
	MVB12	hMVB12A, B	-	-	
ESCRT-II	VPS22/SNF8	EAP30	VPS22 (AT4G27040)	VPS22 (Pp3c10_13560 / Pp6c10_7170)	Pp1s141_146 V6.1
	VPS25	EAP20	VPS25 (AT4G19003)	VPS25 (Pp3c15_1420 / Pp6c15_780)	Pp1s211_155 V6.1
	VPS36	EAP45	VPS36 (AT5G04920)	VPS36 (Pp3c23_21630 / Pp6c23_10660)	Pp1s335_30V 6.1
ESCRT-III core proteins	VPS20	CHMP6	VPS20-1 (AT5G63880) VPS20-2 (AT5G09260)	VPS20 (Pp3c2_32100 / Pp6c2_17850)	Pp1s426_21V 6.1

	SNF7/VP S32	CHMP4A, B, C	SNF7-1 (AT2G19830) SNF7-2 (AT4G29160)	SNF7A (Pp3c3_36260 / Pp6c3_19290) SNF7B (Pp3c4_3530 / Pp6c4_1880) SNF7C (Pp3c12_22980 / Pp6c12_11270) SNF7D (Pp3c8_3630 / Pp6c8_2050)	Pp1s96_36V6 .1 Pp1s287_65V 6.1 Pp1s118_150 V6.1 Pp1s35_247V 6.1
	VPS24	CHMP3	VPS24-1 (AT5G22950) VPS24-2 (AT3G45000)	VPS24A (Pp3c9_19830 / Pp6c9_9920) VPS24B (Pp3c15_14920 / Pp6c15_7680) VPS24C (Pp3c15_15320 / Pp6c15_7860)	Pp1s89_19V6 .1 Pp1s104_70V 6.1 Pp1s104_29V 6.1
	VPS2/DID 4	CHMP2A CHMP2B	VPS2-1 (AT2G06530) VPS2-2 (AT5G44560) VPS2-3 (AT1G03950)	VPS2A (Pp3c23_14890 / Pp6c23_7020) VPS2B (Pp3c3_2240 / Pp6c3_1220) VPS2C (Pp3c13_6090 / Pp6c13_3050) VPS2D (Pp3c13_8060 / Pp6c13_3870) VPS2E (Pp3c3_3880 / Pp6c3_2170)	Pp1s156_55V 6.1 Pp1s1_416V6 .1 Pp1s182_65V 6.1 Pp1s52_141V 6.1 Pp1s1_598V6 .1
ESCRT- III accessor y proteins	DID2	CHMP1A CHMP1B	CHMP1A (AT1G73030) CHMP1B (AT1G17730)	CHMP1A (Pp3c8_6470 / Pp6c8_3480) CHMP1B (Pp3c20_11050 / Pp6c20_6990) CHMP1C (Pp3c10_20850 / Pp6c10_11140) CHMP1D (Pp3c20_20650 / Pp6c20_10870)	Pp1s8_63V6. 1 Pp1s406_15V 6.1 Pp1s67_233V 6.1 Pp1s40_174V 6.1
	VPS60	CHMP5	VPS60-1 (AT3G10640)	VPS60A (Pp3c27_2180 / Pp6c25_3260) VPS60B (Pp3c27_2240 /	Pp1s54_150V 6.1 Pp1s54_155V

			VPS60-2 (AT5G04850)	Pp6c25_3310) VPS60C (Pp3c16_16920 / Pp6c16_13520)	6.1 Pp1s197_29V 6.1
	IST1		ISTL1 (AT1G34220) ISTL2 (AT1G25420) ISTL3 (AT4G35730) ISTL4 (AT4G29440) ISTL5 (AT2G19710) ISTL6 (AT1G13340) ISTL7 (AT4G32350) ISTL8 (AT1G79910) ISTL9 (AT1G52315) ISTL10 (AT2G14830) ISTL11 (AT3G15490) ISTL12 (AT1G51900)	ISTL1A (Pp3c2_13590 / Pp6c2_7700) ISTL1B (Pp3c10_1180 / Pp6c10_340) ISTL1C (Pp3c14_9750 / Pp6c14_5410) ISTL1D (Pp3c1_18360 / Pp6c1_10040)	Pp1s30_79V6 .1 Pp1s95_96V6 .1 Pp1s79_42V6 .1 Pp1s43_42V6 .1
	CMP7	CHMP7	CHMP7 (AT3G62080)	CHMP7 (Pp3c3_29200 / Pp6c3_15360)	Pp1s55_36V6 .1
VPS4 and accessor y proteins	VPS4	VPS4	VPS4/SKD1 (AT2G27600)	VPS4A (Pp3c1_1310 / Pp6c1_790) VPS4B (Pp3c2_35940 / Pp6c2_19890)	Pp1s200_72V 6.1 Pp1s76_197V 6.1
	VTA1	LIP5	LIP5/VTA1 (AT4G26750)	LIP5 (Pp3c12_1420 / Pp6c12_360)	Pp1s128_16V 6.1
Other ESCRT- related proteins	BRO1	ALIX	BRO1/ALIX (AT1G15130)	BRO1A (Pp3c8_14930 / Pp6c8_7310) BRO1B (Pp3c24_13730 / Pp6c24_7130)	Pp1s363_9V6 .1 Pp1s403_53V 6.1
	-	-	FREE1/FYVE1 (AT1G20110)	FYVE1 (Pp3c14_11260 / Pp6c14_6230)	Pp1s36_316V 6.1
	-	-	FYVE4 (AT1G61690)	FYVE4 (Pp3c16_8350 / Pp6c16_5490)	Pp1s194_180 V6.1
	-	-	PROS (AT4G24370)	-	

^a Categorization follows previous reports in (Gao *et al.*, 2017; Gonzalez Solis *et al.*, 2022). Sc, *Saccharomyces cerevisiae*; Hs, *Homo sapiens*; At, *Arabidopsis thaliana*; Pp, *Physcomitrium patens*.

^b Homologs are identified via BLASTing *Arabidopsis* proteins against *P. patens* proteome (version 6.1) in Phytozome database. Gene IDs from version 3.3 and 6.1 are both provided.

Video legends

Video S1. Localization of VPS28A-mNG and Lifeact-mCherry during cell division. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S2. Localization of VPS36-mNG during cell division. Images were captured at 1-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S3. Localization of VPS20-mNG during cell division. Images were captured at 1-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S4. Localization of VPS4A-mNG and Lifeact-mCherry during cell division. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S5. Localization of LIP5-mNG and Lifeact-mCherry during cell division. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S6. Dynamics of VPS4A-mNG at the cell plate. Images were captured at 1.43-second intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S7. Dynamics of LIP5-mNG at the cell plate. Images were captured at 0.5-second intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S8. Dynamics of VPS4A-mNG and RABF1A-mCherry in a caulonemal tip cell. Images were captured at 0.5-second intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S9. Dynamics of VPS4A-mNG and mCherry-RABF2A in a caulonemal tip cell. Images were captured at 0.5-second intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S10. Localization of VPS4A-mNG and RABF1A-mCherry during cell division. Images were captured at 1-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S11. Localization of VPS4A-mNG and mCherry-RABF2A during cell division. Images were captured at 1-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S12. Vacuolar morphology of a caulonemal tip cell in the *vps4a*^{E237Q}-*iox* line. Cells were labeled by GFP- α -tubulin and treated with 1 μ M β -estradiol for 10 minutes. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S13. Mitotic division of a caulonemal tip cell in the *WT* plant. Cells were labeled by GFP- α -tubulin and Histone 2B-RFP. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S14. Mitotic division of a caulonemal tip cell in the *vps4-hm* mutant. Cells were labeled by GFP- α -tubulin and Histone 2B-RFP. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S15. Localization of VPS4A-mNG and mCherry-RABA1D during cell division. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S16. Localization of mCherry-RABA1D during cell division in *WT* plants. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S17. Localization of mCherry-RABA1D during cell division in the *vps4-hm* mutant. This line also expressed a histone reporter (H2B-RFP), which did not interfere with the visualization of RABA1D at the cell plate. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S18. Localization of DRP1A-mCherry during cell division in *WT* plants. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S19. Localization of DRP1A-mCherry during cell division in the *vps4-hm* mutant.

This line also expressed a histone reporter (H2B-RFP), which did not interfere with the visualization of DRP1A at the cell plate. Images were captured at 2-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S20. Localization of AtSKD1-mNG during cell division. This line also expressed a histone reporter (H2B-RFP) and a tubulin reporter (GFP- α -tubulin), which did not interfere with the visualization of AtSKD1 on vesicles and at the cell plate. Images were captured at 1-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

Video S21. Localization of ScVPS4-mNG during cell division This line also expressed a histone reporter (H2B-RFP) and a tubulin reporter (GFP- α -tubulin), which did not interfere with the visualization of ScVPS4 on vesicles and at the cell plate. Images were captured at 1-minute intervals. Playback rate: 10 frames per second. The video should be viewed using Fiji or ImageJ.

References

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